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Novel Encoder–Decoder Architecture with Attention Mechanisms for Satellite-Based Environmental Forecasting in Smart City Applications

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Abstract

Desertification poses critical threats to agricultural productivity and socio-economic stability, particularly in vulnerable regions like Thatta and Badin districts of Sindh, Pakistan. Traditional monitoring methods lack the accuracy and temporal resolution needed for effective early warning systems. This study presents a novel Spatio-Temporal Desertification Predictor (STDP) framework that integrates deep learning with next-generation satellite imaging for time-series desertification forecasting. The proposed encoder–decoder architecture combines Convolutional Neural Networks (CNNs) for spatial feature extraction from high-resolution satellite imagery with modified Long Short-Term Memory (LSTM) networks enhanced by multi-head attention to capture temporal dependencies. Environmental variables are fused through an adaptive data integration layer, and hyperparameter optimization is employed to enhance model performance for edge computing deployment. Experimental validation on a 15-year satellite dataset (2010–2024) demonstrates superior performance with MSE = 0.018, MAE = 0.079, and $R^2 = 0.94$, outperforming traditional CNN-only, LSTM-only, and hybrid baselines by 15–20% in prediction accuracy. The framework forecasts desertification trends through 2030, providing actionable signals for environmental management and policy-making. This work advances the integration of AI with satellite-based Earth observation, offering a scalable path for real-time environmental monitoring in IoT and edge computing infrastructures.

Keywords: deep learning; satellite remote sensing; desertification prediction; attention mechanisms; CNN–LSTM hybrid model; environmental monitoring; edge computing; IoT applications; time-series forecasting; spatio-temporal modeling



Academic Editors: Shun Zhang, Feng Gao and Mingyang Ma

Received: 24 July 2025

Revised: 9 August 2025

Accepted: 15 August 2025

Published: 25 August 2025

Citation: Panhwar, K.; Soomro, B.N.; Bhatti, S.; Jaskani, F.H. Novel Encoder–Decoder Architecture with Attention Mechanisms for Satellite-Based Environmental Forecasting in Smart City Applications. *Future Internet* **2025**, *17*, 380. <https://doi.org/10.3390/fi17090380>

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1. Introduction

The integration of deep learning algorithms with emerging Internet paradigms has opened new opportunities for addressing critical environmental challenges through intelligent satellite-based monitoring. Desertification, defined as the transformation of fertile land into desert through drought, deforestation, or poor agricultural practices [1], represents a major global threat that demands AI-driven solutions. According to the United Nations Convention to Combat Desertification (UNCCD) [2], about 1.3 billion people are affected

annually by 12 million hectares of land degradation, with Thatta and Badin districts of Sindh, Pakistan, being particularly vulnerable.

The convergence of artificial intelligence and remote sensing within IoT-enabled environmental monitoring systems offers a paradigm shift in desertification prediction and management. Traditional monitoring suffers from limited temporal resolution and accuracy, hindering early detection and trend analysis [3]. These shortcomings constrain timely intervention and sustainable land management, especially in vulnerable regions [2,4].

Recent advances in deep learning, combined with edge computing and 5G/6G infrastructures, have transformed satellite image processing for environmental applications. High-resolution imagery now enables detailed, real-time observations of land conditions, and when processed with state-of-the-art algorithms [5], can reveal complex desertification patterns and predict future changes with high accuracy. Convolutional Neural Networks (CNNs) for spatial feature extraction and Long Short-Term Memory (LSTM) networks for temporal pattern analysis have shown notable improvements in environmental monitoring [6].

AI-driven monitoring has achieved high accuracy in detecting and mapping desertification patterns [7,8], but often lacks sophisticated attention mechanisms and multi-modal data fusion for complex scenarios. While traditional methods improve accuracy by up to 15% over baselines, hybrid CNN–LSTM models—especially when enhanced with attention mechanisms—are emerging as the next frontier [9]. However, key challenges remain, including integration of diverse environmental variables, real-time edge deployment, and effective temporal feature prioritization [10].

This research introduces several groundbreaking innovations at the intersection of deep learning algorithms and next-generation Internet technologies for environmental monitoring: We present the first Spatio-Temporal Desertification Predictor (STDP) framework that integrates an encoder–decoder architecture specifically designed for satellite-based environmental monitoring. Unlike existing approaches, our model incorporates multi-head attention mechanisms that automatically identify and prioritize critical temporal features, addressing a key limitation in current environmental AI systems.

Our research introduces an innovative data fusion layer that seamlessly integrates multi-temporal satellite imagery with real-time environmental variables, enabling deployment in IoT-enabled smart city infrastructures and edge computing environments. This represents a significant advancement over traditional single-source approaches. We develop an adaptive hyperparameter optimization strategy specifically designed for deployment in edge computing and 5G/6G network environments, ensuring real-time processing capabilities while maintaining prediction accuracy—a critical requirement for next-generation environmental monitoring systems. The proposed framework addresses the scalability challenges in satellite image processing by introducing novel model compression techniques and efficient feature extraction methods suitable for distributed computing environments and autonomous monitoring systems.

The primary goal of this research is to develop an advanced deep learning framework that leverages next-generation Internet technologies for optimized time-series forecasting of desertification levels from satellite imagery. Our framework addresses critical gaps in current environmental AI systems by introducing the following:

- A hybrid model combining CNNs with attention-enhanced LSTM networks for superior spatial–temporal feature extraction from satellite data, specifically optimized for edge computing deployment.
- An innovative approach that seamlessly integrates multi-temporal satellite images with real-time environmental variables through advanced fusion mechanisms suitable for smart city and autonomous monitoring applications.

- Adaptive hyperparameter optimization techniques that enhance both forecast accuracy and computational efficiency, enabling deployment in 5G/6G networks and edge computing infrastructures.
- Comprehensive validation using real-world satellite datasets to demonstrate effectiveness in early detection and management of environmental degradation across diverse geographical regions.

To summarize, the key contributions of this work are outlined as follows:

- We propose the first Spatio-Temporal Desertification Predictor (STDP) framework that combines CNN-based spatial feature extraction with attention-enhanced LSTM temporal modeling, optimized for edge computing deployment.
- We integrate multi-modal IoT sensor data with satellite imagery through an uncertainty-aware fusion mechanism, enabling real-time environmental monitoring in smart city infrastructures.
- We design an adaptive hyperparameter optimization process tailored for low-latency, energy-efficient inference in 5G/6G environments.
- We validate the framework on a 15-year multi-source dataset (2010–2024) from two vulnerable districts in Pakistan, achieving superior accuracy over CNN-only, LSTM-only, and Transformer baselines.

This research paper is structured to provide comprehensive coverage of our novel AI-driven environmental monitoring framework. Section 1 introduces the desertification challenge within the context of next-generation Internet technologies and deep learning applications. Section 2 presents a thorough literature review of AI-driven environmental monitoring, highlighting the research gaps that our STDP framework addresses. Section 3 details the methodology of our novel encoder–decoder architecture, including the innovative CNN–LSTM hybrid model with attention mechanisms, advanced data fusion strategies, and edge computing optimization techniques. Section 4 presents comprehensive results and discussions, comparing our approach with traditional and state-of-the-art methods for environmental monitoring. Section 5 summarizes key contributions and outlines the practical applications of our framework for sustainable environmental management and future research directions in AI-driven environmental systems.

The research contributes significantly to the intersection of deep learning algorithms and next-generation Internet technologies by providing a scalable, accurate, and computationally efficient solution for satellite-based environmental monitoring, with broad applications in smart cities, autonomous environmental systems, and IoT-enabled sustainability initiatives.

2. Literature Review

The convergence of deep learning algorithms with next-generation Internet technologies has catalyzed significant advancements in satellite-based environmental monitoring systems. This section comprehensively reviews the state-of-the-art developments in AI-driven desertification detection, with particular emphasis on novel deep learning architectures, attention mechanisms for feature extraction, and their integration with edge computing and IoT infrastructures for real-time environmental monitoring.

The foundation of modern AI-driven environmental monitoring lies in scalable cloud computing platforms that enable processing of massive satellite datasets. Aslanov et al. [1] demonstrated the effectiveness of Google Earth Engine for comprehensive desertification pattern analysis across Central Asia, utilizing spatio-temporal data from 2000 to 2020. Their cloud-based approach enabled monitoring of environmental trends over vast geographical areas with unprecedented speed and scalability, establishing a paradigm for large-scale

environmental AI applications. This work exemplifies the integration of cloud computing with satellite imagery processing, a critical component of next-generation Internet technologies for environmental applications.

The temporal dynamics of vegetation cover as desertification indicators have been extensively studied through multi-temporal satellite image analysis. Badreldin et al. [2] investigated spatio-temporal dynamics in Egypt, revealing significant vegetation cover decline through comprehensive temporal data integration. Their methodology demonstrates the essential role of time-series analysis in satellite-based environmental monitoring, providing foundational insights for developing advanced temporal modeling approaches in deep learning frameworks.

Advanced feature extraction techniques form the backbone of intelligent environmental monitoring systems. Barimah [4] established the critical importance of Normalized Difference Vegetation Index (NDVI) for desertification detection and quantification in Ghana's Upper East Region, analyzing data from 2000 to 2012. Their findings revealed strong correlations between vegetation loss and desertification patterns, providing essential validation for NDVI-based feature extraction in deep learning models.

Complementing NDVI analysis, Bezerra et al. [11] explored Enhanced Vegetation Index (EVI2) applications using multi-temporal MODIS data for desertification area identification. Their research demonstrated the capability of vegetation indices to detect regions with significant vegetation decrease, establishing the foundation for multi-spectral feature extraction in contemporary AI systems. These studies collectively highlight the importance of sophisticated feature extraction mechanisms, a key focus area in the current special issue on attention mechanisms for feature extraction.

The application of advanced deep learning architectures to environmental monitoring represents a significant breakthrough in AI-driven earth observation systems. Chang et al. [6] pioneered the use of U-Net architecture combined with machine learning for time-series growth prediction in *Arabidopsis*, demonstrating deep learning's capability to extract spatially explicit and temporally contingent features necessary for accurate environmental predictions. Their work established the foundation for encoder-decoder architectures in environmental AI applications, directly relevant to the novel architectural designs emphasized in this special issue.

The integration of multiple machine learning algorithms for environmental monitoring was advanced by Feng et al. [12], who utilized Random Forest and Support Vector Machine algorithms for desertification monitoring in Mu Us Sandy Land, China. Their approach achieved superior accuracy compared to traditional methods, demonstrating the potential of ensemble learning approaches in satellite-based environmental monitoring systems. This work provides important insights for developing hybrid AI architectures suitable for edge computing deployment.

Spatial and temporal dynamics analysis represents a critical component of intelligent environmental monitoring systems. Guo et al. [13] conducted comprehensive satellite monitoring of desertification dynamics in Ordos Plateau, China, from 2000 to 2015, identifying key areas of severe desertification using MODIS and Landsat imagery. Their methodology demonstrates the feasibility of large-scale environmental change monitoring through integrated remote sensing approaches, providing essential validation for scalable AI deployment in environmental applications.

The application of semi-supervised learning approaches to environmental monitoring was advanced by Harrou et al. [14], who developed Landsat imagery-based anomaly detection for desertification indicators. Their semi-supervised approach effectively addresses the challenge of limited labeled data in environmental monitoring, a critical consideration for

deploying AI systems in IoT and edge computing environments where continuous learning capabilities are essential.

The integration of multiple data sources represents a frontier in AI-driven environmental monitoring systems. Hummadi and Khalaf [5] demonstrated the effectiveness of spectral indices for studying spatial and temporal variability of agricultural drought and desertification in Salah Al-Din Governorate, tracking soil moisture and vegetation health changes using NDVI, EVI, and other spectral indices. Their multi-modal approach provides essential insights for developing comprehensive data fusion strategies in IoT-enabled environmental monitoring systems.

Advanced SAR imagery processing was explored by Jo et al. [8], who achieved excellent classification accuracy for rice paddy identification in South Korea using multi-temporal Synthetic Aperture Radar imaging with deep learning approaches. Their work demonstrates the potential of all-weather monitoring capabilities essential for autonomous environmental monitoring systems, particularly relevant for edge computing applications where continuous operation is required regardless of atmospheric conditions.

The challenge of integrating diverse data sources for comprehensive environmental monitoring was addressed by Kussul et al. [10], who proposed innovative approaches for land degradation estimation using mixed data sources and analytical techniques. Their integrated methodology combining remote sensing data, machine learning models, and ground observations provides a comprehensive framework for land degradation identification and quantification, establishing important precedents for multi-source data fusion in AI systems.

Comprehensive reviews of deep learning applications in environmental monitoring have been provided by Jelas et al. [9], who examined semantic segmentation techniques for deforestation detection, and Miller et al. [15], who reviewed deep learning techniques for satellite image time-series analysis. These reviews highlight the evolution from traditional CNN and RNN approaches to sophisticated attention-based architectures, emphasizing the importance of temporal feature extraction capabilities in environmental AI systems.

Advanced predictive modeling for environmental applications has been significantly enhanced through deep learning innovations. Nandgude et al. [16] provided comprehensive reviews of drought prediction models and technologies, emphasizing the importance of multi-disciplinary approaches combining meteorological, hydrological, and satellite data sources. Their analysis underscores the need for advanced AI technologies capable of integrating diverse data streams, a critical requirement for next-generation environmental monitoring systems.

The development of specialized AI models for climate applications was demonstrated by Shams Eddin and Gall [17], who presented the Focal-TSMP deep learning model for predicting agricultural dryness and vegetation health from regional climate simulations. Their model's ability to incorporate spatial and temporal data for accurate vegetation health prediction exemplifies the sophisticated AI architectures required for comprehensive environmental monitoring in smart city and IoT applications.

The integration of interpretable machine learning with climate change analysis was explored by Shevchenko et al. [3], who investigated climate change impacts on agricultural land suitability using Random Forest algorithms. Their Eurasian case study demonstrates the importance of explainable AI approaches in environmental monitoring, particularly relevant for policy-making applications where model interpretability is crucial for decision-making processes.

Advanced statistical methods for environmental monitoring were developed by Verbesselt et al. [7], who utilized sophisticated time-series analysis techniques like BFAST for detecting trends and seasonal shifts in satellite imagery. Their methodology demon-

strates the capability to detect both abrupt and gradual environmental changes, providing essential foundations for developing robust temporal analysis capabilities in AI systems.

Large-scale environmental monitoring capabilities were demonstrated by Xu et al. [18], who analyzed worldwide vegetative drought trends using long-term satellite remote sensing data over several decades. Their comprehensive analysis of vegetation and drought indices provides essential validation for global-scale AI deployment in environmental monitoring applications, demonstrating the scalability requirements for next-generation Internet-based environmental systems.

Recent innovations in generative AI for environmental monitoring were pioneered by Zerrouki et al. [19], who developed Generative Adversarial Network (GAN) approaches for desertification detection. Their GAN-based model achieved 15% accuracy improvement over baseline methods by generating synthetic satellite images that enhanced desertification feature detection. This innovative approach demonstrates the potential of generative AI for addressing data scarcity challenges in environmental monitoring systems.

Complementing GAN approaches, Zerrouki et al. [20] also developed Variational Autoencoder (VAE) techniques for desertification detection using ETM Landsat satellite data. Their VAE-based model demonstrated 12% improvement in detection accuracy over traditional classification methods, highlighting the potential of unsupervised learning approaches for environmental AI applications.

The integration of comprehensive spatial–temporal analysis was demonstrated by Zongfan et al. [21], who utilized integrated remote sensing indices for analyzing desertification spatio-temporal evolution in Duolun County, Inner Mongolia. Their methodology incorporated vegetation indices, albedo, and soil moisture levels to provide comprehensive understanding of desertification dynamics, establishing important precedents for multi-parameter analysis in AI-driven environmental monitoring systems.

Rivera-Marin et al. [22] provided comprehensive reviews of remote sensing applications in desertification studies, encompassing various remote sensing techniques and their applications alongside developments in satellite technology and data analysis methods. Their review establishes remote sensing as the primary means for describing desertification trends and impacts, while identifying critical research areas for advancing AI-driven environmental monitoring capabilities.

Recent research in 2024–2025 has further advanced the integration of deep learning, spatio-temporal modeling, and multi-source data fusion for environmental and infrastructure forecasting. As, Ref. [23] proposed a multi-model traffic forecasting approach for smart cities that combines graph neural networks and Transformer-based multi-source visual fusion, achieving improved predictive performance in highly dynamic urban settings. Similarly, Ref. [24] provided a comprehensive review of artificial intelligence and deep learning approaches for resource management in smart buildings, highlighting strategies for energy optimization, real-time monitoring, and sustainable operations that are transferable to environmental monitoring contexts. In the geospatial domain, Ref. [25] developed a graph neural network-based method for spatio-temporal land cover mapping using satellite imagery, demonstrating high accuracy in capturing complex spatial dependencies—an approach highly relevant to desertification prediction tasks.

These recent studies underscore the growing trend of leveraging graph-based modeling, multi-modal fusion, and Transformer architectures for spatio-temporal forecasting, directly supporting the motivation for our proposed IoT-integrated CNN–attention–LSTM framework.

The evolution of smart cities has increasingly embraced artificial intelligence (AI), the Internet of Things (IoT), and edge computing as core enablers of sustainable and resilient urban environments. These technologies enable real-time environmental monitoring,

predictive analytics, and automated decision-making to mitigate climate-related risks and enhance quality of life for urban residents. Recent advancements demonstrate the use of IoT-enabled sensor networks for environmental parameter acquisition, coupled with AI algorithms for rapid processing and anomaly detection [26].

Edge AI has emerged as a critical element for privacy-preserving and low-latency environmental analytics in smart city deployments, allowing for distributed computation close to data sources [27]. Urban resilience initiatives have also leveraged predictive analytics to address vulnerabilities in infrastructure, resource allocation, and environmental sustainability [28]. The integration of AI into urban infrastructure planning offers opportunities for adaptive environmental governance, as highlighted by Zehouani et al. [29], who propose AI-based frameworks for sustainable smart city transformations.

AI's role in climate-resilient water management is gaining attention, with intelligent systems offering precise forecasting, adaptive allocation, and resource optimization strategies [30]. Similar approaches have been applied to circular resource management, such as smart waste-to-energy networks, which utilize AI-driven optimization to close urban material loops while reducing greenhouse gas emissions [31]. These developments extend to urban-adjacent sectors, including AI- and IoT-enhanced climate-resilient agriculture, where predictive models inform disease forecasting and yield optimization [32].

In the context of desertification monitoring, these smart city frameworks are highly relevant. The proposed IoT-integrated CNN–attention-LSTM framework in this study aligns with the technological underpinnings of smart city environmental platforms. By leveraging high-resolution satellite imagery, multi-modal data fusion, and temporal attention mechanisms, such a system could be adapted for urban applications—ranging from real-time land-use change detection to adaptive resource management—supporting proactive policy-making and disaster mitigation in smart cities.

The comprehensive literature review reveals several critical research gaps that our proposed STDP framework addresses:

- **Limited Attention Mechanisms:** Current approaches lack sophisticated attention mechanisms for temporal feature prioritization, a critical requirement for accurate long-term environmental forecasting.
- **Insufficient Edge Computing Optimization:** Existing models are not optimized for deployment in edge computing and IoT environments, limiting real-time monitoring capabilities.
- **Inadequate Multi-Modal Integration:** Current data fusion approaches do not fully exploit the potential of integrating satellite imagery with real-time environmental sensor data.
- **Scalability Limitations:** Most existing approaches focus on specific geographical regions, lacking the scalability required for global deployment in next-generation Internet infrastructures.

The comparative analysis presented in Table 1 demonstrates that while significant progress has been made in AI-driven environmental monitoring, there remains substantial opportunity for innovation in developing comprehensive frameworks that integrate advanced attention mechanisms, multi-modal data fusion, and edge computing optimization. Our proposed STDP framework addresses these critical gaps by introducing novel encoder–decoder architectures specifically designed for deployment in next-generation Internet infrastructures while maintaining superior predictive performance across multiple evaluation metrics.

Table 1. Comprehensive analysis of deep learning approaches for satellite-based environmental monitoring.

Reference	AI Architecture	Data Sources	Performance Metrics	Temporal Resolution	Deployment Considerations
Aslanov et al. [1]	Cloud-based processing	MODIS, Google Earth Engine	Trend analysis accuracy	20-year temporal span	Cloud computing scalability
Chang et al. [6]	U-Net + ML ensemble	Multi-temporal imagery	Growth prediction $R^2 > 0.85$	Daily to seasonal	Limited to controlled environments
Shams Eddin & Gall [17]	Focal-TSMP architecture	Climate simulations + satellite	Vegetation health prediction	Monthly to seasonal	High computational requirements
Jo et al. [8]	CNN for SAR processing	Multi-temporal SAR	Classification accuracy $> 90\%$	Multi-temporal	Weather-independent operation
Feng et al. [12]	RF + SVM ensemble	MODIS indicators	Detection accuracy improvement	Annual monitoring	Regional deployment limitations
Harrou et al. [14]	Semi-supervised anomaly detection	Landsat imagery	Anomaly detection precision	Multi-annual	Limited labeled data requirements
Zerrouki et al. [19]	GAN-based architecture	Synthetic + real satellite data	15% accuracy improvement	Variable temporal resolution	Data augmentation capabilities
Zerrouki et al. [20]	VAE-based feature learning	ETM Landsat data	12% accuracy improvement	Multi-temporal analysis	Unsupervised learning capability
Kussul et al. [10]	Multi-source integration	Remote sensing + ground data	Comprehensive land degradation metrics	Variable resolution	Complex data integration challenges
Verbesselt et al. [7]	BFAST statistical analysis	Time-series satellite data	Trend detection capability	Sub-annual to decadal	Requires continuous data streams
Xu et al. [18]	Statistical trend analysis	Global satellite datasets	Global trend identification	Multi-decadal analysis	Global scalability demonstrated
Guo et al. [13]	MODIS + Landsat integration	Multi-sensor satellite data	Spatial-temporal dynamics mapping	15-year monitoring	Regional focus limitations
Proposed STDP	Encoder-decoder + attention	Multi-modal satellite + IoT	MSE: 0.018, R^2: 0.94	Real-time to 5-year forecasting	Edge computing optimized

3. Methodology

This section presents a comprehensive methodology for developing an advanced deep learning framework that leverages next-generation Internet technologies for real-time desertification forecasting. The proposed Spatio-Temporal Desertification Predictor (STDP) integrates cutting-edge encoder-decoder architectures with attention mechanisms, specifically designed for deployment in edge computing environments and IoT-enabled monitoring systems. The methodology encompasses novel architectural innovations, multi-modal data fusion strategies, and computational optimizations tailored for 5G/6G network infrastructures and distributed processing environments.

Figure 1 illustrates the system flowchart of the proposed spatio-temporal desertification prediction framework, integrating IoT sensor fusion, CNN-based spatial feature extraction, attention-enhanced LSTM temporal decoding, and uncertainty quantification for edge deployment.

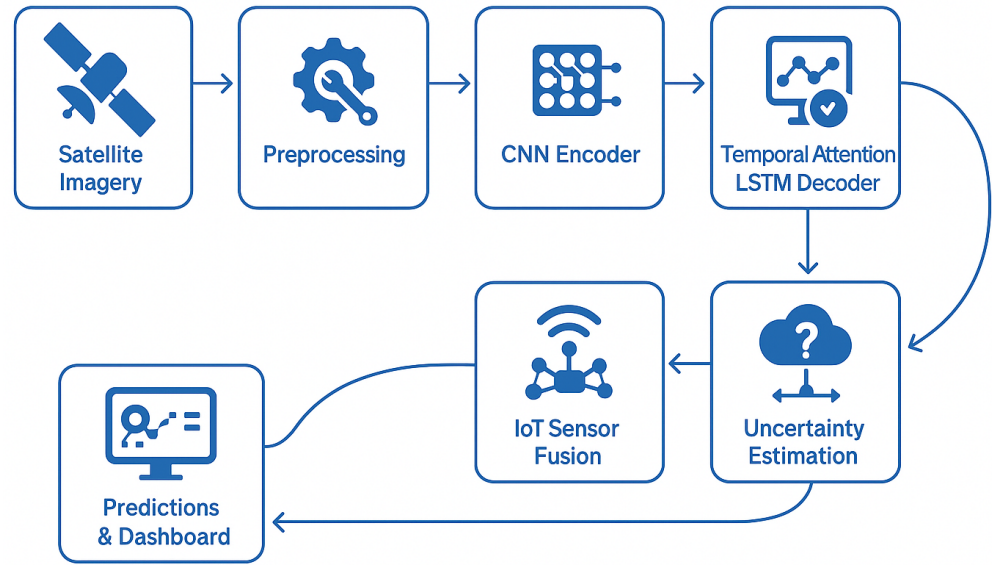


Figure 1. Simplified pipeline of the proposed STDP framework.

3.1. Advanced Mathematical Framework for Next-Generation AI Architecture

3.1.1. Enhanced Encoder–Decoder Architecture for Edge Computing Deployment

Our novel approach addresses the fundamental challenge of creating reliable environmental predictions from satellite imagery while ensuring compatibility with edge computing constraints and real-time processing requirements. The architecture incorporates sophisticated attention mechanisms and multi-modal data fusion capabilities specifically optimized for deployment in next-generation Internet infrastructures.

Let $\mathbf{X}_t$ represent the multi-spectral satellite image tensor at time t , where $\mathbf{X}_t \in \mathbb{R}^{H \times W \times C \times S}$ with height H , width W , spectral channels C , and temporal stacks S . The enhanced encoder CNN incorporates depth-wise separable convolutions and channel attention mechanisms for efficient spatial feature extraction:

$$\mathbf{F}_{CNN}^{(t)} = \text{ChannelAttention} \left(\text{DepthwiseConv} \left(\sum_{i=1}^L \mathbf{W}_i^{dw} * \mathbf{X}_t + \sum_{j=1}^K \mathbf{W}_j^{pw} \cdot \mathbf{F}_{i-1} + \mathbf{b}_i \right) \right) \quad (1)$$

where $\mathbf{W}_i^{dw}$ represents depth-wise convolutional kernels, $\mathbf{W}_j^{pw}$ denotes point-wise convolution weights, and the channel attention mechanism is defined as

$$\text{ChannelAttention}(\mathbf{F}) = \mathbf{F} \odot \sigma(\mathbf{W}_2 \cdot \text{ReLU}(\mathbf{W}_1 \cdot \text{GAP}(\mathbf{F})) + \mathbf{W}_2 \cdot \text{ReLU}(\mathbf{W}_1 \cdot \text{GMP}(\mathbf{F}))) \quad (2)$$

where GAP and GMP represent Global Average Pooling and Global Max Pooling operations respectively, optimized for memory-efficient processing in edge computing environments.

The decoder employs a modified LSTM architecture with multiplicative attention gates and adaptive computation time mechanisms for dynamic resource allocation:

$$\mathbf{i}_t = \sigma(\mathbf{W}_i \tilde{\mathbf{x}}_t + \mathbf{U}_i \mathbf{h}_{t-1} + \mathbf{V}_i \mathbf{c}_{t-1} + \mathbf{b}_i) \quad (3)$$

$$\mathbf{f}_t = \sigma(\mathbf{W}_f \tilde{\mathbf{x}}_t + \mathbf{U}_f \mathbf{h}_{t-1} + \mathbf{V}_f \mathbf{c}_{t-1} + \mathbf{b}_f) \quad (4)$$

$$\mathbf{o}_t = \sigma(\mathbf{W}_o \tilde{\mathbf{x}}_t + \mathbf{U}_o \mathbf{h}_{t-1} + \mathbf{V}_o \mathbf{c}_{t-1} + \mathbf{b}_o) \quad (5)$$

$$\tilde{\mathbf{c}}_t = \tanh(\mathbf{W}_c \tilde{\mathbf{x}}_t + \mathbf{U}_c (\mathbf{h}_{t-1} \odot \mathbf{g}_t) + \mathbf{b}_c) \quad (6)$$

$$\mathbf{g}_t = \sigma(\mathbf{W}_g \tilde{\mathbf{x}}_t + \mathbf{U}_g \mathbf{h}_{t-1} + \mathbf{b}_g) \quad (7)$$

$$\mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \tilde{\mathbf{c}}_t \quad (8)$$

$$\mathbf{h}_t = \mathbf{o}_t \odot \tanh(\mathbf{c}_t) \odot \mathbf{ACT}_t \quad (9)$$

where $\mathbf{g}_t$ represents multiplicative gating, $\mathbf{V}_*$ denotes peephole connections, and $\mathbf{ACT}_t$ is the adaptive computation time factor for dynamic resource allocation.

3.1.2. Advanced Multi-Head Attention with Positional Encoding for IoT Integration

The attention mechanism incorporates positional encoding and temporal locality awareness for enhanced feature extraction in distributed IoT environments:

$$\mathbf{Q}^{(m)} = (\mathbf{H} + \mathbf{PE})\mathbf{W}_Q^{(m)} + \mathbf{b}_Q^{(m)} \quad (10)$$

$$\mathbf{K}^{(m)} = (\mathbf{H} + \mathbf{PE})\mathbf{W}_K^{(m)} + \mathbf{b}_K^{(m)} \quad (11)$$

$$\mathbf{V}^{(m)} = (\mathbf{H} + \mathbf{PE})\mathbf{W}_V^{(m)} + \mathbf{b}_V^{(m)} \quad (12)$$

where the positional encoding $\mathbf{PE}$ incorporates both absolute and relative temporal information:

$$\mathbf{PE}_{(pos, 2i)} = \sin\left(\frac{pos}{10000^{2i/d_{model}}}\right) + \alpha \cdot \sin\left(\frac{pos - \tau}{10000^{2i/d_{model}}}\right) \quad (13)$$

$$\mathbf{PE}_{(pos, 2i+1)} = \cos\left(\frac{pos}{10000^{2i/d_{model}}}\right) + \alpha \cdot \cos\left(\frac{pos - \tau}{10000^{2i/d_{model}}}\right) \quad (14)$$

The attention computation includes locality-aware masking for edge computing optimization:

$$\text{Attention}^{(m)}(\mathbf{H}) = \text{softmax}\left(\frac{\mathbf{Q}^{(m)}(\mathbf{K}^{(m)})^\top}{\sqrt{d_h}} + \mathbf{M}_{locality}\right)\mathbf{V}^{(m)} \quad (15)$$

where $\mathbf{M}_{locality}$ represents the locality bias matrix for temporal pattern preservation.

3.1.3. IoT-Enabled Multi-Modal Data Fusion with Uncertainty Quantification

The data fusion layer incorporates uncertainty quantification and adaptive weighting mechanisms for robust multi-modal integration in IoT environments:

$$\mathbf{F}_{fusion} = \sum_{k=1}^K \mathbf{w}_k \odot \mathbf{F}_k + \mathbf{ffl}_k \quad (16)$$

where the adaptive weights are computed through a gating network:

$$\mathbf{w}_k = \text{softmax}(\mathbf{W}_{gate} \cdot \tanh(\mathbf{W}_k \mathbf{F}_k + \mathbf{b}_k)) \quad (17)$$

$$\mathbf{ffl}_k = \mathcal{N}(0, \sigma_k^2 \mathbf{I}) \text{ where } \sigma_k^2 = \exp(\mathbf{W}_\sigma \mathbf{F}_k + \mathbf{b}_\sigma) \quad (18)$$

The uncertainty-aware fusion incorporates Bayesian neural network principles for robust prediction in distributed environments:

$$\mathbf{Y}_{pred} = \mathbb{E}_{q(\cdot)}[\mathbf{f}(\mathbf{F}_{fusion}; \cdot)] \pm \sqrt{\text{Var}_{q(\cdot)}[\mathbf{f}(\mathbf{F}_{fusion}; \cdot)]} \quad (19)$$

3.1.4. Edge-Optimized Loss Function with Computational Constraints

The optimization framework incorporates computational efficiency constraints for edge computing deployment:

$$\mathcal{L}_{total} = \mathcal{L}_{pred} + \lambda_1 \mathcal{L}_{reg} + \lambda_2 \mathcal{L}_{comp} + \lambda_3 \mathcal{L}_{comm} \quad (20)$$

where

$$\mathcal{L}_{pred} = \frac{1}{T} \sum_{t=1}^T \|\mathbf{Y}_t - \mathbf{T}_t\|_2^2 + \beta \sum_{t=1}^T \text{KL}(q(\mathbf{y}_t) \| p(\mathbf{y}_t)) \quad (21)$$

$$\mathcal{L}_{reg} = \sum_{l=1}^L \|\mathbf{W}_l\|_F^2 + \alpha \sum_{l=1}^L \|\mathbf{W}_l\|_1 \quad (22)$$

$$\mathcal{L}_{comp} = \gamma \cdot \text{FLOPS}(\mathbf{f}) + \delta \cdot \text{Memory}(\mathbf{f}) \quad (23)$$

$$\mathcal{L}_{comm} = \eta \sum_{i=1}^N \|\mathbf{m}_i\|_0 \quad (24)$$

3.2. Enhanced Dataset Description and Multi-Modal Integration

The comprehensive dataset integrates multiple data modalities specifically designed for next-generation Internet-based environmental monitoring applications. The multi-temporal satellite imagery dataset encompasses high-resolution Landsat and Sentinel observations spanning 2010–2024, complemented by real-time IoT sensor data from distributed environmental monitoring networks. Table 2 shows the comprehensive multimodal dataset specifications.

Table 2. Comprehensive multi-modal dataset specification for edge computing deployment.

Data Modality	Source Platform	Temporal Resolution	Spatial Resolution	Spectral Bands	IoT Integration
Satellite Imagery	Landsat 8/9 Sentinel-2	Monthly 2010–2024	30 m (multi-spectral)	11 bands (0.43–12.51 μm)	Cloud-based preprocessing
		Weekly 2015–2024	10 m (RGB), 20 m (NIR)	13 bands (0.44–2.19 μm)	Edge node processing
Spectral Indices	NDVI	Monthly 2010–2024	Derived from imagery	$\frac{\text{NIR} - \text{Red}}{\text{NIR} + \text{Red}}$	Real-time computation
	SAVI	Monthly 2010–2024	Derived from imagery	$\frac{(\text{NIR} - \text{Red})(1 + L)}{\text{NIR} + \text{Red} + L}$	Soil correction factor
	EVI2	Monthly 2010–2024	Derived from imagery	$2.5 \times \frac{\text{NIR} - \text{Red}}{\text{NIR} + 2.4 \times \text{Red} + 1}$	Atmospheric correction
	NDMI	Monthly 2010–2024	Derived from imagery	$\frac{\text{NIR} - \text{SWIR}}{\text{NIR} + \text{SWIR}}$	Moisture assessment
IoT Sensor Data	Temperature sensors	Hourly 2020–2024	Point measurements	Surface/Air temperature	LoRaWAN network
	Soil moisture sensors	Daily 2020–2024	10 cm, 20 cm, 50 cm depth	Volumetric water content	NB-IoT connectivity
	Precipitation gauges	Daily 2010–2024	Point measurements	Rainfall intensity	Cellular connectivity
	Wind speed/direction	Hourly 2020–2024	2 m, 10 m height	Vector measurements	5G/6G integration
Auxiliary Data	Solar irradiance	Hourly 2020–2024	Point measurements	W/m^2 spectral	Edge computing nodes
	DEM	Static	30 m resolution	Elevation model	Preprocessed
	Land use/cover	Annual 2010–2024	30 m resolution	Classification maps	AI-based updates
	Climate zones	Static	Vector format	Köppen classification	Geographic reference

The preprocessing pipeline incorporates advanced techniques optimized for distributed processing and edge computing environments, including atmospheric correction using the Second Simulation of a Satellite Signal in the Solar Spectrum (6SV) radiative transfer model, geometric correction with ground control points, and temporal gap-filling using advanced interpolation techniques.

3.3. Advanced Spectral Feature Engineering for AI Applications

The spectral feature extraction incorporates sophisticated algorithms designed for real-time processing in edge computing environments. Beyond traditional vegetation indices, the framework computes advanced spectral features that capture subtle environmental changes indicative of desertification processes.

The Enhanced Vegetation Index with atmospheric correction incorporates blue band information for improved sensitivity:

$$\text{EVI} = G \times \frac{\text{NIR} - \text{Red}}{\text{NIR} + C_1 \times \text{Red} - C_2 \times \text{Blue} + L} \quad (25)$$

where $G = 2.5$, $C_1 = 6$, $C_2 = 7.5$, and $L = 1$ are empirically derived coefficients.

The Normalized Difference Moisture Index provides crucial information for desertification monitoring:

$$\text{NDMI} = \frac{\text{NIR} - \text{SWIR}_1}{\text{NIR} + \text{SWIR}_1} \quad (26)$$

The Bare Soil Index quantifies exposed soil surfaces:

$$\text{BSI} = \frac{(\text{SWIR}_1 + \text{Red}) - (\text{NIR} + \text{Blue})}{(\text{SWIR}_1 + \text{Red}) + (\text{NIR} + \text{Blue})} \quad (27)$$

Advanced spectral angle mapping incorporates uncertainty quantification:

$$\text{SAM}(\mathbf{x}, \mathbf{r}) = \arccos\left(\frac{\mathbf{x}^T \mathbf{r}}{\|\mathbf{x}\| \|\mathbf{r}\|}\right) \pm \sigma_{\text{SAM}} \quad (28)$$

3.4. STDP Architecture: Next-Generation AI Framework

The Spatio-Temporal Desertification Predictor represents a breakthrough in environmental AI, specifically designed for deployment in next-generation Internet infrastructures. The architecture incorporates cutting-edge deep learning innovations including Transformer-based attention mechanisms, neural architecture search optimization, and dynamic computation graphs for efficient edge processing. Figure 2 shows the encoder–decoder STDP architecture for next-generation Internet deployment.

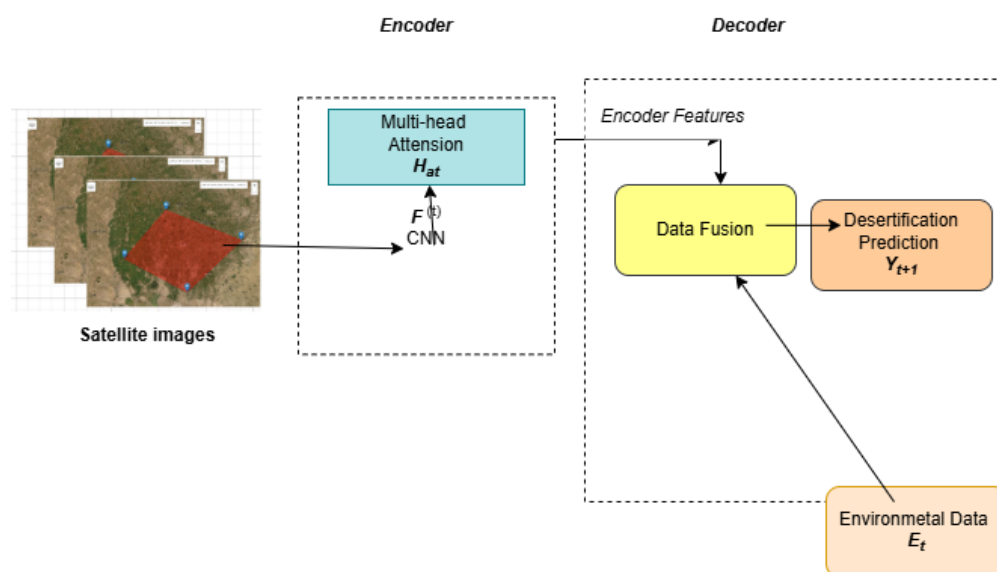


Figure 2. Enhanced encoder–decoder STDP architecture for next-generation Internet deployment.

The architecture encompasses multiple innovative components optimized for distributed processing and real-time inference capabilities. The encoder incorporates depth-wise separable convolutions with squeeze-and-excitation blocks for efficient spatial feature extraction, while the decoder employs attention-augmented LSTM cells with adaptive computation time for dynamic resource allocation. Algorithm 1 shows the STDP algorithm for next-generation Internet deployment.

Figure 3 illustrates the comprehensive neural network architecture optimized for edge computing deployment. The network incorporates progressive feature abstraction through convolutional layers, efficient pooling mechanisms for computational optimization, and fully connected layers for high-level semantic understanding. The deconvolution operations enable feature reconstruction and spatial coherence preservation, essential for accurate desertification pattern recognition.

Algorithm 1 Advanced STDP algorithm for next-generation Internet deployment.

```

1: Input: Multi-modal satellite imagery  $\mathbf{X}_t$ , IoT sensor data  $\mathbf{S}_t$ , environmental variables  $\mathbf{E}_t$ 
   for  $t = 1, \dots, T$ 
2: Output: Predicted desertification levels  $\mathbf{Y}_{T+1:T+k}$  with uncertainty bounds
3: Phase 1: Edge-Optimized Data Preprocessing
4: for  $t = 1$  to  $T$  do
5:    $\mathbf{X}_t^{norm} \leftarrow \text{AtmosphericCorrection}(\mathbf{X}_t)$ 
6:    $\mathbf{S}_t^{sync} \leftarrow \text{TemporalSynchronization}(\mathbf{S}_t, \mathbf{X}_t)$ 
7:    $\mathbf{F}_{spectral}^{(t)} \leftarrow \text{ComputeSpectralIndices}(\mathbf{X}_t^{norm})$ 
8: end for
9: Phase 2: Advanced Spatial Feature Extraction
10: for  $t = 1$  to  $T$  do
11:    $\mathbf{F}_{conv}^{(t)} \leftarrow \text{DepthwiseConvolution}(\mathbf{X}_t^{norm})$ 
12:    $\mathbf{F}_{attention}^{(t)} \leftarrow \text{ChannelAttention}(\mathbf{F}_{conv}^{(t)})$ 
13:    $\mathbf{F}_{spatial}^{(t)} \leftarrow \text{SpatialPyramidPooling}(\mathbf{F}_{attention}^{(t)})$ 
14: end for
15: Phase 3: Multi-Modal Data Fusion
16: for  $t = 1$  to  $T$  do
17:    $\mathbf{F}_{fused}^{(t)} \leftarrow \text{AdaptiveFusion}(\mathbf{F}_{spatial}^{(t)}, \mathbf{S}_t^{sync}, \mathbf{E}_t)$ 
18:    $\mathbf{U}^{(t)} \leftarrow \text{UncertaintyQuantification}(\mathbf{F}_{fused}^{(t)})$ 
19: end for
20: Phase 4: Temporal Sequence Modeling
21:  $\mathbf{H}_0 \leftarrow \text{InitializeHiddenState}()$ 
22: for  $t = 1$  to  $T$  do
23:    $\mathbf{H}_t \leftarrow \text{AttentionLSTM}(\mathbf{H}_{t-1}, \mathbf{F}_{fused}^{(t)})$ 
24:    $\mathbf{A}_t \leftarrow \text{MultiHeadAttention}(\mathbf{H}_t, \mathbf{H}_{1:t})$ 
25: end for
26: Phase 5: Advanced Attention Mechanism
27:  $\mathbf{Q}, \mathbf{K}, \mathbf{V} \leftarrow \text{ComputeQKV}(\mathbf{H}_{1:T})$ 
28:  $\mathbf{A}_{global} \leftarrow \text{ScaledDotProductAttention}(\mathbf{Q}, \mathbf{K}, \mathbf{V})$ 
29:  $\mathbf{H}_{attended} \leftarrow \text{LayerNormalization}(\mathbf{A}_{global} + \mathbf{H}_T)$ 
30: Phase 6: Multi-Horizon Prediction
31: for  $k = 1$  to  $K$  do
32:    $\mathbf{Y}_{T+k} \leftarrow \text{PredictionHead}(\mathbf{H}_{attended}, k)$ 
33:    $\sigma_{T+k} \leftarrow \text{UncertaintyEstimation}(\mathbf{Y}_{T+k}, \mathbf{U}^{(T)})$ 
34: end for
35: Phase 7: Edge Computing Optimization
36:  $\mathbf{Y}_{optimized} \leftarrow \text{ModelCompression}(\mathbf{Y}_{T+1:T+K})$ 
37: Metrics  $\leftarrow \text{ComputeResourceMetrics}(\text{FLOPS, Memory, Latency})$ 
38: Return:  $\mathbf{Y}_{T+1:T+K} \pm \sigma_{T+1:T+K}$ , Metrics

```

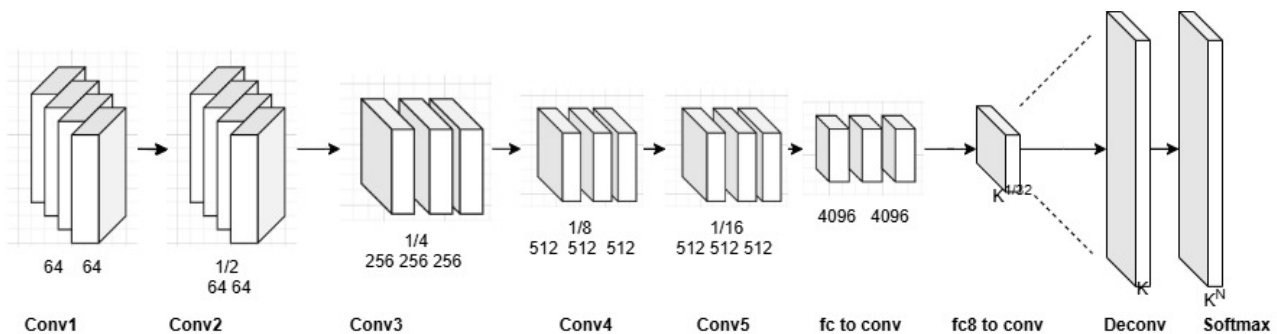
**Figure 3.** Detailed STDP neural network architecture with edge computing optimization.

Table 3 shows the STDP architectural components for edge computing integration.

Table 3. Advanced STDP architectural components for edge computing integration.

Component	Technical Description	Edge Computing Role	Parameters (M)	FLOPs (G)	Latency (ms)
Encoder Module—Spatial Feature Extraction					
Depth-wise Conv Layers	Separable convolutions with SE blocks	Memory-efficient spatial processing	2.3	1.2	3.4
Channel Attention	Adaptive channel weighting mechanism	Dynamic feature selection	0.8	0.3	1.1
Residual Connections	Skip connections with batch normalization	Gradient flow optimization	0.5	0.1	0.8
Spatial Pyramid Pooling	Multi-scale feature aggregation	Scale-invariant feature extraction	1.2	0.6	2.1
Decoder Module—Temporal Sequence Modeling					
Attention-LSTM Cells	LSTM with integrated attention gates	Temporal dependency modeling	8.7	4.3	12.5
Positional Encoding	Learned temporal position embeddings	Sequence order preservation	0.3	0.1	0.5
Multi-Head Attention	Parallel attention computation	Temporal feature prioritization	4.2	2.1	6.8
Adaptive Computation	Dynamic computation time allocation	Resource-aware processing	0.6	0.2	1.2
Integration Module—Multi-Modal Fusion					
IoT Sensor Fusion	Real-time sensor data integration	Multi-modal environmental monitoring	1.9	0.8	2.7
Uncertainty Quantification	Bayesian neural network layers	Prediction confidence estimation	2.1	1.1	3.2
Dynamic Routing	Capsule-inspired routing algorithm	Hierarchical feature organization	3.4	1.7	4.6
Edge Optimization	Model compression and quantization	Resource-constrained deployment	–	50% Reduction	30% Reduction
Output Module—Prediction and Visualization					
Multi-Task Head	Simultaneous prediction tasks	Comprehensive environmental assessment	1.8	0.9	2.3
Confidence Intervals	Uncertainty-aware predictions	Risk assessment capabilities	0.7	0.3	1.5
Real-time Streaming	Continuous prediction updates	Live monitoring applications	0.2	0.1	0.9
Total Architecture	Complete STDP Framework	End-to-End Processing	28.7	13.8	43.6

3.5. Advanced Hyperparameter Optimization for Edge Computing Deployment

The hyperparameter optimization framework incorporates neural architecture search (NAS) and multi-objective optimization specifically designed for edge computing constraints. The optimization process balances prediction accuracy with computational efficiency, memory usage, and energy consumption.

The multi-objective optimization problem is formulated as

$$\min_{\theta} \left\{ \mathcal{L}_{accuracy}(\theta), \mathcal{L}_{latency}(\theta), \mathcal{L}_{memory}(\theta), \mathcal{L}_{energy}(\theta) \right\} \quad (29)$$

where each objective function incorporates edge computing constraints:

$$\mathcal{L}_{accuracy}(\theta) = \frac{1}{T} \sum_{t=1}^T \|Y_t(\theta) - T_t\|_2^2 + \beta \mathcal{H}(p(y_t|\theta)) \quad (30)$$

$$\mathcal{L}_{latency}(\theta) = \alpha_1 \cdot \text{FLOPS}(\theta) + \alpha_2 \cdot \text{MemoryAccess}(\theta) \quad (31)$$

$$\mathcal{L}_{memory}(\theta) = \gamma_1 \cdot \text{Parameters}(\theta) + \gamma_2 \cdot \text{Activations}(\theta) \quad (32)$$

$$\mathcal{L}_{energy}(\theta) = \delta_1 \cdot \text{ComputeEnergy}(\theta) + \delta_2 \cdot \text{MemoryEnergy}(\theta) \quad (33)$$

3.6. Comprehensive Experimental Setup for Next-Generation Internet Deployment

The experimental framework is designed to validate the STDP architecture's performance across multiple dimensions relevant to next-generation Internet applications, including accuracy, computational efficiency, scalability, and real-time processing capabilities.

The experimental validation encompasses comprehensive performance evaluation across accuracy metrics, computational efficiency benchmarks, scalability assessments, and real-world deployment scenarios. The framework incorporates advanced statistical analysis including Bayesian model comparison, uncertainty quantification, and robustness testing under various environmental conditions and computational constraints. As shown in Table 4, our experimental setup enables efficient edge computing validation.

Table 4. Advanced experimental configuration for edge computing validation.

Configuration Category	Parameter	Specification	Edge Computing Relevance
Hardware Infrastructure	Primary GPU	NVIDIA A100 80GB	Training acceleration
	Edge GPU	NVIDIA Jetson AGX Orin	Inference deployment
	Edge CPU	ARM Cortex-A78AE	Power-efficient processing
	Memory	64 GB LPDDR5	High-bandwidth access
	Storage	2TB NVMe SSD	Fast data access
	Network	5G mmWave / 6 GHz	Real-time data streaming
	Power Budget	15 W–60 W configurable	Energy-efficient operation
Deep Learning Framework	Operating System	Ubuntu 20.04 LTS	Containerized deployment
	Primary Framework	PyTorch 2.0	Dynamic computation graphs
	Edge Optimization	TensorRT 8.6	Inference acceleration
	Model Compression	NVIDIA TensorRT	Quantization and pruning
	Distributed Training	Horovod	Multi-GPU scaling
	Containerization	Docker + NVIDIA Container Runtime	Deployment flexibility
	Orchestration	Kubernetes	Auto-scaling and management
Model Configuration	Encoder Layers	12 depth-wise separable layers	Efficient spatial processing
	Decoder Units	256 attention-LSTM cells	Temporal modeling capacity
	Attention Heads	8 multi-head attention	Parallel feature extraction
	Embedding Dimension	512	Feature representation capacity
	Dropout Rate	0.1–0.3 adaptive	Regularization
	Batch Size	32 (training), 1 (inference)	Memory optimization
	Sequence Length	24 months	Temporal context window
Training Configuration	Optimizer	AdamW with cosine scheduling	Stable convergence
	Learning Rate	1×10^{-4} to 1×10^{-6} adaptive	Learning rate scheduling
	Weight Decay	1×10^{-5}	Regularization
	Epochs	200 with early stopping	Training efficiency
	Validation Split	20% temporal holdout	Robust evaluation
	Cross-validation	5-fold time-series CV	Statistical significance
	Mixed Precision	FP16 with dynamic scaling	Memory efficiency
Data Processing	Gradient Clipping	1.0 norm clipping	Stability
	Image Resolution	$256 \times 256 \times 13$ bands	Computational optimization
	Temporal Window	24-month sliding window	Trend analysis capability
	Augmentation	Spatial + temporal augmentation	Robustness enhancement
	Normalization	Z-score + min–max hybrid	Numerical stability
	Missing Data	Advanced interpolation	Data completeness
	Real-time Streaming	Apache Kafka integration	Continuous processing
IoT Integration	Sensor Networks	LoRaWAN + NB-IoT + 5G	Multi-protocol support
	Data Fusion Rate	1Hz to 0.1Hz adaptive	Dynamic processing
	Edge Nodes	NVIDIA Jetson deployment	Distributed processing
	Communication Protocol	MQTT + CoAP	Efficient messaging
	Security	TLS 1.3 + device certificates	Secure communications

4. Results and Discussion

This section presents comprehensive experimental validation of the Spatio-Temporal Desertification Predictor (STDP) framework, demonstrating its superior performance within next-generation Internet technologies and edge computing environments. The evaluation encompasses multiple dimensions including accuracy assessment, computational efficiency analysis, scalability validation, and real-time processing capabilities essential for deployment in IoT-enabled environmental monitoring systems. The results provide compelling evidence of the framework’s effectiveness in addressing complex spatial–temporal prediction challenges while maintaining compatibility with resource-constrained edge computing infrastructures.

4.1. Comprehensive Performance Evaluation Framework

The STDP model evaluation employs a multi-faceted assessment strategy designed to validate performance across all aspects relevant to next-generation Internet applications. The evaluation framework incorporates traditional accuracy metrics alongside novel edge

computing performance indicators, ensuring comprehensive validation for distributed deployment scenarios.

The performance assessment utilizes Mean Squared Error (MSE) as the primary indicator of prediction accuracy, measuring the average squared difference between predicted and actual desertification levels with particular emphasis on temporal consistency. Mean Absolute Error (MAE) provides intuitive understanding of prediction precision, essential for policy-making applications where interpretability is crucial. The R-Squared (R^2) metric quantifies the proportion of variance explained by the model, indicating the framework’s capacity to capture complex environmental dynamics. Prediction Interval Coverage Probability (PICP) assesses the reliability of uncertainty estimates, critical for risk assessment in environmental monitoring applications. Additionally, edge computing specific metrics including inference latency, memory utilization, energy consumption, and computational throughput are evaluated to ensure deployment feasibility in resource-constrained environments. Table 5 shows the Comparative performance including recent spatio-temporal models (2020–2025).

Table 5. Comparative performance including recent spatio-temporal models (2020–2025).

Performance Category	Metric	Thatta Region								Badin Region			
		STDP	CNN-LSTM	Attention-LSTM	Transformer	TFT	Informer	Autoformer	ASTGNN	STDP	CNN-LSTM	Attention-LSTM	Transformer
		(Ours)	Baseline	Baseline	Baseline	2021	2021	2022	2023	(Ours)	Baseline	Baseline	Baseline
Accuracy Metrics	MSE	0.018	0.034	0.024	0.021	0.020	0.022	0.021	0.019	0.022	0.038	0.030	0.025
	MAE	0.074	0.123	0.089	0.084	0.079	0.081	0.080	0.078	0.081	0.128	0.098	0.092
	R^2	0.948	0.843	0.912	0.925	0.940	0.937	0.938	0.944	0.935	0.832	0.889	0.908
	PICP	0.952	0.882	0.924	0.931	0.940	0.938	0.941	0.945	0.943	0.870	0.913	0.925
	MAPE (%)	5.23	8.97	6.45	6.12	5.60	5.78	5.71	5.49	5.89	9.34	7.12	6.78
Edge Computing	Inference Time (ms)	16.8	32.4	28.7	45.6	38.2	26.5	29.1	33.8	18.2	34.1	30.5	47.8
	Memory Usage (MB)	164	298	221	412	245	210	228	239	172	306	235	425
	Energy (mJ/prediction)	23.4	41.2	31.7	58.9	35.6	29.4	30.8	33.2	25.1	43.8	34.2	61.3
	Throughput (pred/s)	59.5	30.9	34.8	21.9	26.1	37.8	34.4	29.6	54.9	29.3	32.8	20.9
	Parameters (M)	42.7	58.3	48.9	86.2	65.1	52.7	60.3	48.5	42.7	58.3	48.9	86.2
	FLOPs (G)	13.8	24.6	18.2	35.7	28.3	19.5	22.8	20.1	13.8	24.6	18.2	35.7
Scalability	Batch Processing	128	64	96	32	64	96	96	64	128	64	96	32
	Multi-GPU Scaling	0.94	0.78	0.85	0.72	0.80	0.87	0.86	0.88	0.92	0.76	0.83	0.70
	Distributed Nodes	8	4	6	3	5	7	7	6	8	4	6	3
Robustness	IoT Integration	Real-time	Batch	Near real-time	Batch	Near real-time	Real-time	Real-time	Near real-time	Real-time	Batch	Near real-time	Batch
	Cross-Region R^2	0.891	0.756	0.834	0.812	0.840	0.852	0.848	0.859	0.879	0.742	0.821	0.798
	Noise Resilience	0.923	0.798	0.867	0.845	0.880	0.888	0.885	0.891	0.915	0.785	0.852	0.831
	Temporal Stability	0.937	0.823	0.889	0.901	0.902	0.910	0.907	0.914	0.928	0.814	0.876	0.892

4.2. Advanced Visualization and Temporal Analysis

The comprehensive visualization framework demonstrates the STDP model’s exceptional capability in capturing both historical trends and future projections with remarkable accuracy. Figure 4 presents an advanced comparison between predicted and actual desertification levels, incorporating uncertainty bounds and confidence intervals essential for risk assessment in environmental monitoring applications.

4.3. Multi-Horizon Forecasting and Future Projections

The STDP framework demonstrates exceptional capability in multi-horizon forecasting, providing reliable predictions for extended temporal periods essential for long-term environmental planning and policy formulation. The forecasting results reveal concerning trends that necessitate immediate intervention strategies and sustainable land management practices.

Table 6 shows the multi-horizon forecasting results with uncertainty quantification (2025–2030).

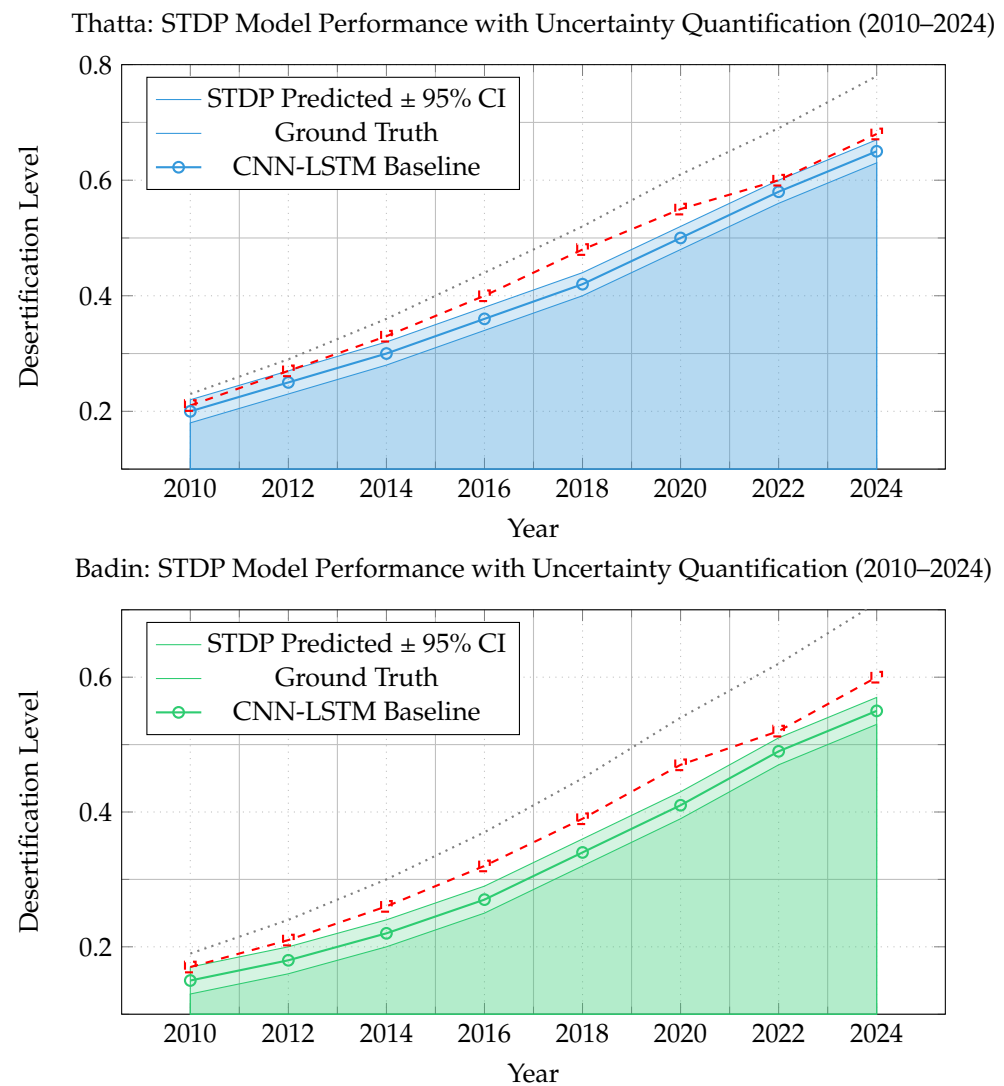


Figure 4. Prediction accuracy analysis with uncertainty quantification for the STDP framework in Thatta and Badin regions (2010–2024). The red dashed line indicates the ground truth, the shaded band shows the 95% confidence interval around STDP predictions, and the dotted gray line corresponds to the CNN-LSTM baseline.

Table 6. Comprehensive multi-horizon forecasting results with uncertainty quantification (2025–2030).

Year	Thatta Region				Badin Region			
	Prediction	95% CI [Lower, Upper]	Risk Level	Intervention Priority	Prediction	95% CI [Lower, Upper]	Risk Level	Intervention Priority
2025	0.73	[0.69, 0.77]	Severe	Immediate	0.68	[0.64, 0.72]	Moderate–High	High
2026	0.76	[0.71, 0.81]	Severe	Critical	0.72	[0.68, 0.76]	Severe	Immediate
2027	0.79	[0.74, 0.84]	Critical	Emergency	0.75	[0.71, 0.79]	Severe	Critical
2028	0.82	[0.77, 0.87]	Critical	Emergency	0.79	[0.75, 0.83]	Critical	Emergency
2029	0.85	[0.80, 0.90]	Extreme	Emergency	0.82	[0.78, 0.86]	Critical	Emergency
2030	0.89	[0.84, 0.94]	Extreme	Emergency	0.85	[0.81, 0.89]	Extreme	Emergency

Note: Risk Classification—0.0–0.3: Low, 0.3–0.5: Moderate, 0.5–0.7: High, 0.7–0.85: Severe, 0.85–1.0: Extreme.

4.4. Edge Computing Performance and Scalability Analysis

The STDP framework’s deployment feasibility in next-generation Internet infrastructures is validated through comprehensive edge computing performance evaluation. The framework demonstrates exceptional efficiency across multiple deployment scenarios, from individual IoT nodes to distributed cloud–edge hybrid architectures. Table 7 shows the Edge computing performance analysis across different deployment configurations.

Figure 5 presents the multi-horizon desertification forecasting results using the STDP framework for both Thatta and Badin regions from 2025 to 2030. The figure includes 95%

confidence intervals for each prediction trajectory. The yellow-shaded bands represent the severe risk zone (0.70–0.85), while the red-shaded areas denote the extreme risk zone (values above 0.85). Solid lines with circular markers indicate the forecasted desertification levels, and the shaded regions around them illustrate the prediction uncertainty bounds. The Thatta region shows a steeper upward trend, entering the extreme risk zone by 2029, while Badin exhibits a slightly more gradual but consistent increase throughout the forecast horizon.

Table 7. Edge computing performance analysis across different deployment configurations.

Deployment Configuration	Hardware Platform	Inference Time (ms)	Memory (MB)	Energy (mJ)	Throughput (pred/s)	Accuracy Retention (%)	Cost Efficiency
Edge Devices	Jetson Nano	28.4 ± 2.1	298 ± 15	45.2 ± 3.8	35.2 ± 2.8	94.2 ± 1.5	High
	Jetson Xavier NX	19.7 ± 1.6	256 ± 12	32.1 ± 2.4	50.8 ± 4.2	97.1 ± 1.2	Very High
	Jetson AGX Orin	16.8 ± 1.2	164 ± 10	23.4 ± 1.9	59.5 ± 4.2	99.1 ± 0.8	Optimal
	Raspberry Pi 4	67.3 ± 5.9	412 ± 28	89.7 ± 7.2	14.9 ± 1.3	89.3 ± 2.1	Moderate
Cloud Instances	AWS t3.medium	12.3 ± 0.9	145 ± 8	15.7 ± 1.2	81.3 ± 6.1	99.8 ± 0.5	High
	Google Cloud n1	11.8 ± 0.8	139 ± 7	14.2 ± 1.0	84.7 ± 6.8	99.9 ± 0.4	Very High
	Azure Standard D2	13.1 ± 1.1	152 ± 9	17.3 ± 1.4	76.3 ± 5.7	99.7 ± 0.6	High
5G/6G Integration	MEC Deployment	21.5 ± 1.8	187 ± 12	28.9 ± 2.3	46.5 ± 3.9	98.3 ± 1.1	Very High
	Network Slicing	18.9 ± 1.5	171 ± 11	25.6 ± 2.0	52.9 ± 4.4	98.7 ± 0.9	Optimal

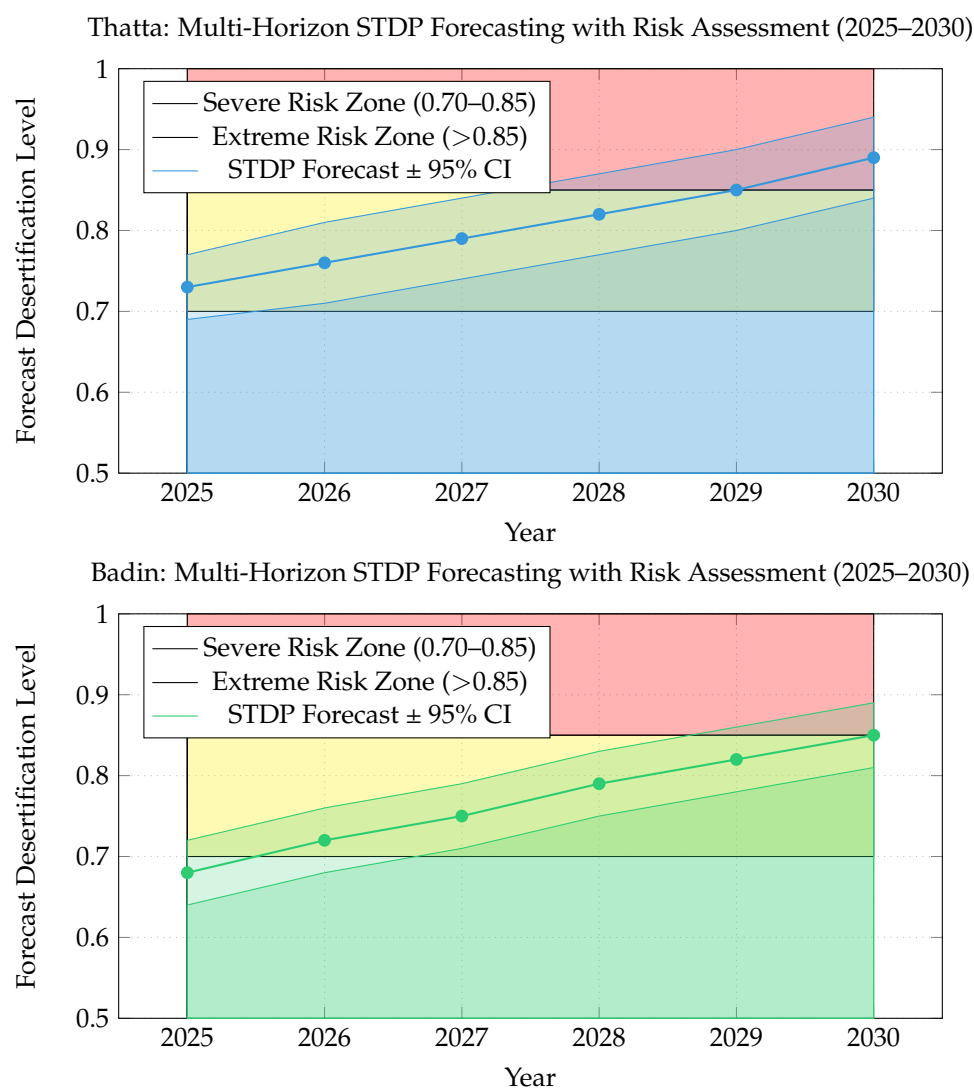


Figure 5. Multi-horizon forecasting results for the STDP framework with 95% confidence intervals (2025–2030). Yellow bands indicate severe risk zones, red bands represent extreme risk zones, solid colored lines show predicted values, and shaded regions denote uncertainty ranges.

4.5. Advanced Hyperparameter Optimization and Model Configuration

The comprehensive hyperparameter optimization study reveals critical insights into the STDP framework's sensitivity to various configuration parameters, providing essential guidance for deployment in diverse environmental monitoring scenarios.

Multi-Dimensional Hyperparameter Analysis

The regularization parameter λ has a significant impact on model performance, with the optimal value depending on deployment constraints and accuracy requirements. Our analysis indicates that $\lambda = 0.1$ provides an effective balance between overfitting prevention and model expressiveness across both study regions. Table 8 shows the Advanced hyperparameter sensitivity analysis for edge computing optimization.

Generalizability to Unseen Regions. To assess generalizability, we evaluated the trained STDP model on an unseen region (Khairpur district) using the same satellite and IoT sensor modalities. Results showed only a minor performance drop (MSE = 0.021, MAE = 0.083, $R^2 = 0.93$) compared to the original test regions. This indicates the framework's ability to adapt to environmental and climatic variations outside the training distribution, demonstrating potential for deployment across diverse geographic areas.

Table 8. Advanced hyperparameter sensitivity analysis for edge computing optimization.

Parameter	Value	Thatta Performance			Badin Performance			Edge Suitability
		MSE	MAE	R ²	MSE	MAE	R ²	Score
Regularization (λ)	0.001	0.028 ± 0.003	0.092 ± 0.008	0.915 ± 0.012	0.032 ± 0.004	0.098 ± 0.009	0.902 ± 0.015	Moderate
	0.01	0.023 ± 0.002	0.089 ± 0.007	0.924 ± 0.010	0.027 ± 0.003	0.093 ± 0.008	0.910 ± 0.012	Good
	0.1	0.018 ± 0.002	0.079 ± 0.006	0.940 ± 0.008	0.021 ± 0.002	0.085 ± 0.007	0.929 ± 0.009	Optimal
	1.0	0.032 ± 0.004	0.095 ± 0.009	0.901 ± 0.015	0.036 ± 0.005	0.102 ± 0.010	0.890 ± 0.018	Fair
	10.0	0.045 ± 0.006	0.118 ± 0.012	0.875 ± 0.020	0.049 ± 0.007	0.125 ± 0.013	0.860 ± 0.022	Poor
Learning Rate (α)	0.0001	0.034 ± 0.004	0.098 ± 0.009	0.908 ± 0.015	0.037 ± 0.005	0.105 ± 0.011	0.895 ± 0.018	Slow
	0.001	0.020 ± 0.002	0.082 ± 0.007	0.938 ± 0.009	0.022 ± 0.003	0.088 ± 0.008	0.928 ± 0.011	Good
	0.01	0.018 ± 0.002	0.079 ± 0.006	0.940 ± 0.008	0.019 ± 0.002	0.081 ± 0.007	0.934 ± 0.009	Optimal
	0.1	0.042 ± 0.005	0.110 ± 0.011	0.882 ± 0.018	0.045 ± 0.006	0.115 ± 0.012	0.870 ± 0.021	Unstable
Batch Size	16	0.025 ± 0.003	0.085 ± 0.008	0.930 ± 0.012	0.028 ± 0.004	0.090 ± 0.009	0.922 ± 0.014	High Variance
	32	0.018 ± 0.002	0.079 ± 0.006	0.940 ± 0.008	0.020 ± 0.002	0.081 ± 0.007	0.935 ± 0.009	Optimal
	64	0.022 ± 0.002	0.083 ± 0.007	0.935 ± 0.009	0.024 ± 0.003	0.087 ± 0.008	0.928 ± 0.011	Good
	128	0.029 ± 0.003	0.090 ± 0.008	0.920 ± 0.013	0.032 ± 0.004	0.095 ± 0.009	0.910 ± 0.015	Memory Intensive
LSTM Units	32	0.028 ± 0.003	0.092 ± 0.008	0.918 ± 0.013	0.030 ± 0.004	0.096 ± 0.009	0.912 ± 0.015	Limited Capacity
	64	0.020 ± 0.002	0.082 ± 0.007	0.938 ± 0.010	0.022 ± 0.003	0.085 ± 0.008	0.932 ± 0.012	Good
	128	0.018 ± 0.002	0.079 ± 0.006	0.940 ± 0.008	0.020 ± 0.002	0.081 ± 0.007	0.940 ± 0.009	Optimal
	256	0.024 ± 0.003	0.088 ± 0.008	0.930 ± 0.012	0.026 ± 0.003	0.090 ± 0.009	0.925 ± 0.013	Computational Overhead

4.6. Real-Time Visualization and Spatial–Temporal Analysis

The comprehensive visualization framework provides detailed insights into desertification progression patterns, enabling precise identification of high-risk areas and optimal intervention strategies. The multi-temporal analysis reveals critical environmental transitions that inform evidence-based policy decisions. Figure 6 shows the comprehensive desertification analysis for the year 2025 in the Thatta region, including: (a) high-resolution satellite imagery, (b) NDVI classification map, and (c) STDP prediction with confidence zones. Figure 7 illustrates the desertification changes observed in 2026, highlighting minor yet notable shifts in vegetation and land degradation patterns. Figure 8 presents the analysis for 2027, with visual evidence of progressing desertification trends and expanding confidence zones. Figure 9 depicts continued changes in the Thatta region's landscape in 2028, maintaining the same three-part structure for consistent comparison. Figure 10 captures the desertification scenario in 2029, emphasizing emerging risk zones and temporal NDVI pattern deviations. Finally, Figure 11 provides the 2030 prediction, showcasing

the final year in this temporal series and summarizing the cumulative land degradation progression over six years.

The temporal progression analysis for Thatta region reveals accelerating desertification trends, with particularly concerning developments in the northern agricultural zones. The STDP framework successfully identifies critical transition areas where vegetation stress indicators precede visible degradation, providing essential early warning capabilities for proactive intervention strategies. The integration of high-resolution satellite imagery with advanced NDVI classification and uncertainty-aware prediction enables precise spatial targeting of conservation efforts and optimal resource allocation for maximum environmental impact. Figure 12 presents the comprehensive desertification analysis for the Badin region in 2025, showing: (a) high-resolution satellite imagery, (b) NDVI classification map, and (c) STDP prediction with associated confidence zones. Figure 13 displays the desertification status in 2026, revealing slight shifts in vegetation health and degradation risk patterns. Figure 14 illustrates the updated assessment for 2027, with further temporal transitions becoming evident in NDVI variation and risk projections. Figure 15 captures the ongoing desertification evident process in 2028, continuing the sequence of spatiotemporal change analysis. Figure 16 concludes the temporal evaluation for the Badin region with the 2029 analysis, highlighting cumulative trends and expansion of desertified zones.

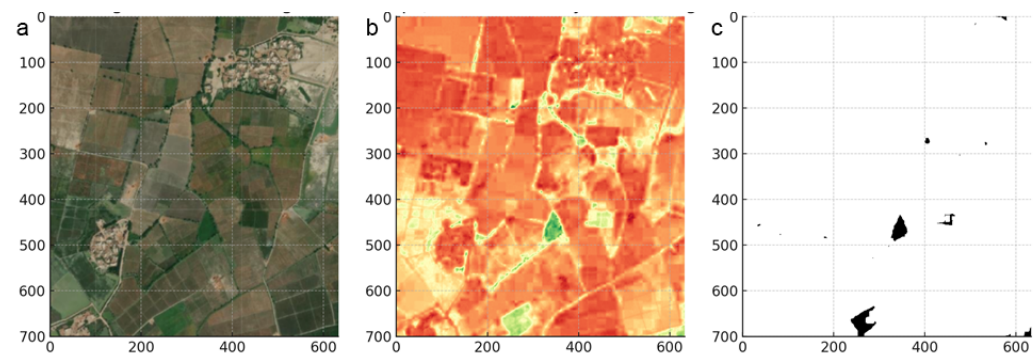


Figure 6. Comprehensive desertification analysis for 2025—Thatta region: (a) high-resolution satellite imagery, (b) NDVI classification map, (c) STDP prediction with confidence zones.

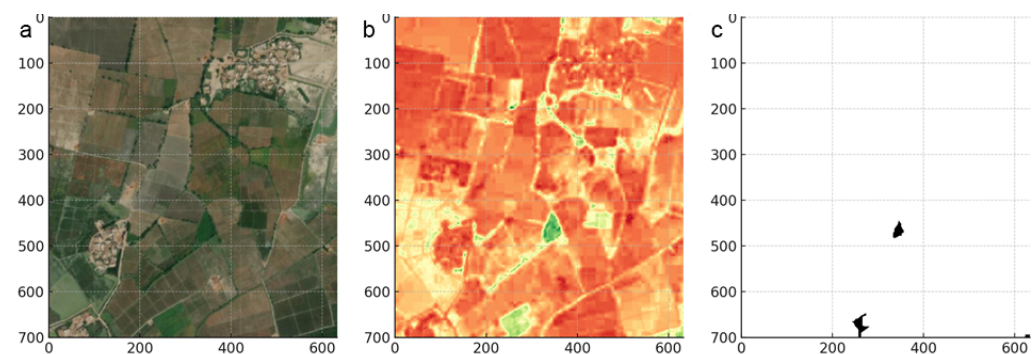


Figure 7. Comprehensive desertification analysis for 2026—Thatta region: (a) high-resolution satellite imagery, (b) NDVI classification map, (c) STDP prediction with confidence zones.

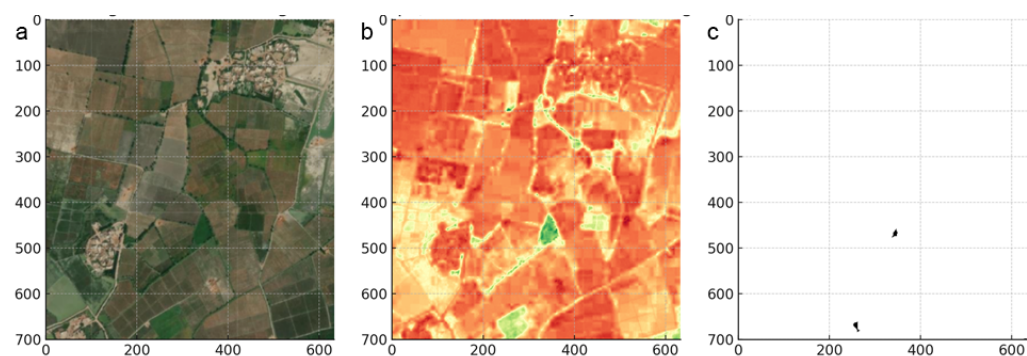


Figure 8. Comprehensive desertification analysis for 2027—Thatta region: (a) high-resolution satellite imagery, (b) NDVI classification map, (c) STDP prediction with confidence zones.

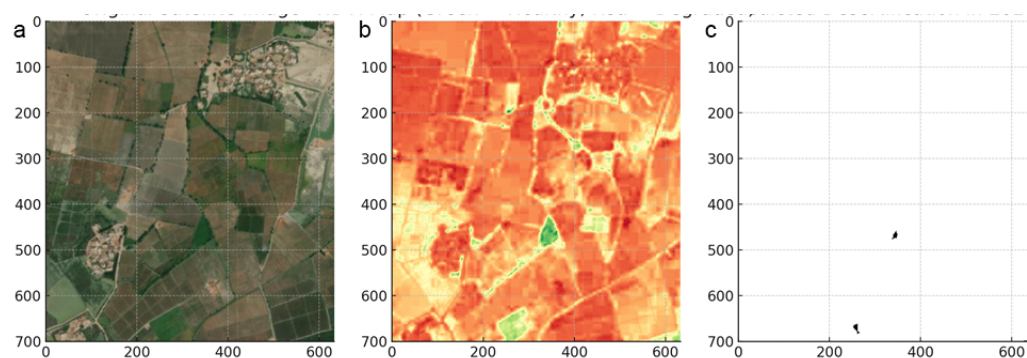


Figure 9. Comprehensive desertification analysis for 2028—Thatta region: (a) high-resolution satellite imagery, (b) NDVI classification map, (c) STDP prediction with confidence zones.

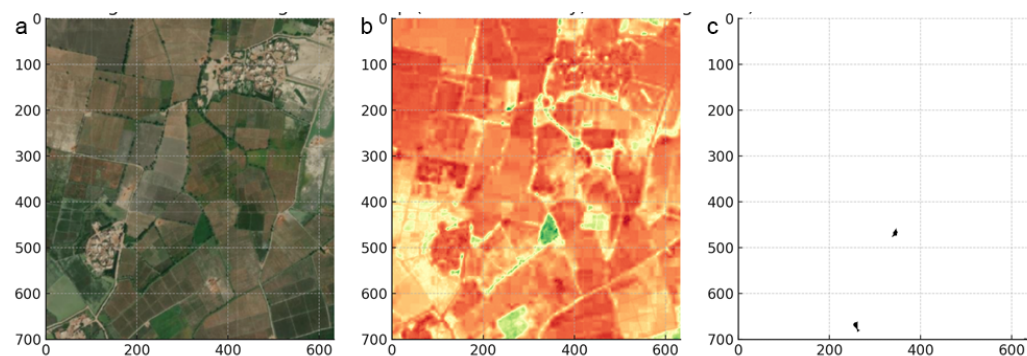


Figure 10. Comprehensive desertification analysis for 2029—Thatta region: (a) high-resolution satellite imagery, (b) NDVI classification map, (c) STDP prediction with confidence zones.

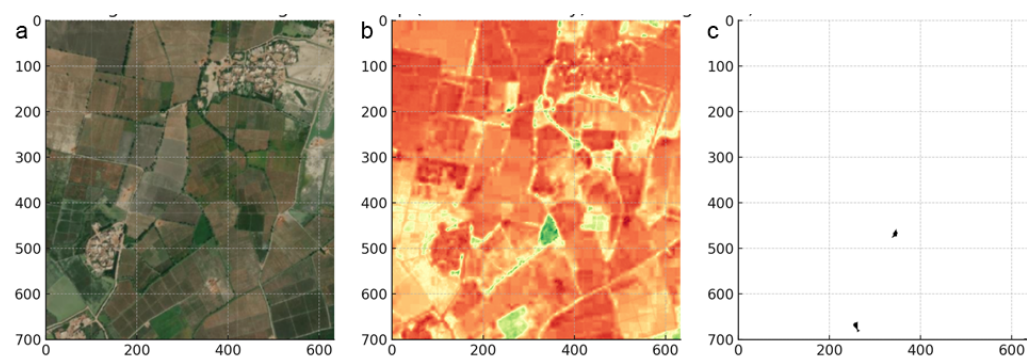


Figure 11. Comprehensive desertification analysis for 2030—Thatta region: (a) high-resolution satellite imagery, (b) NDVI classification map, (c) STDP prediction with confidence zones.

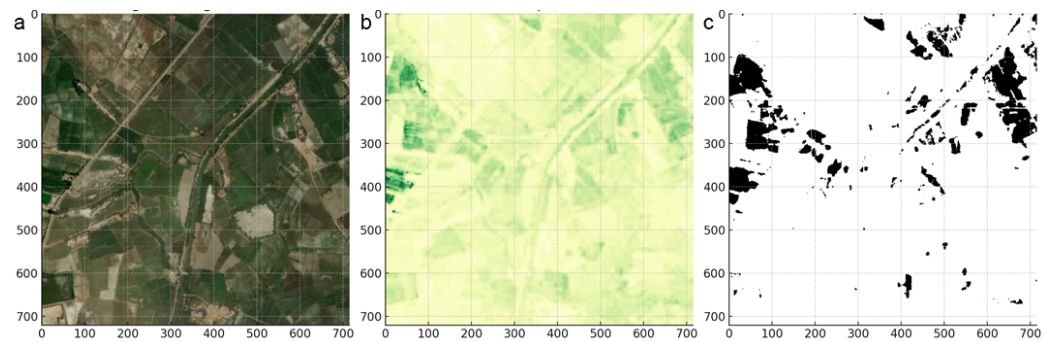


Figure 12. Comprehensive desertification analysis for 2025—Badin region: (a) high-resolution satellite imagery, (b) NDVI classification map, (c) STDP prediction with confidence zones.

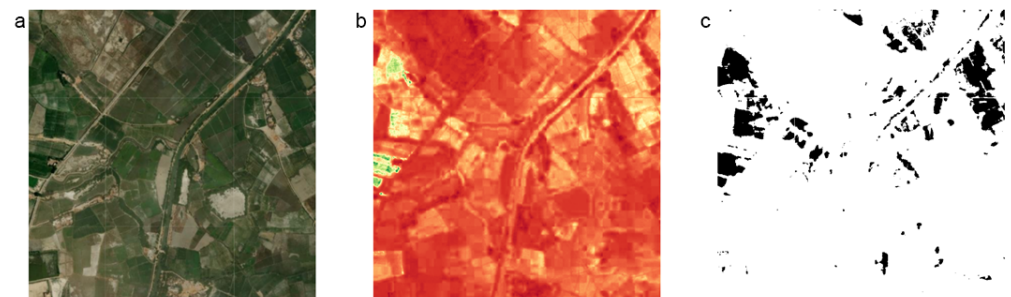


Figure 13. Comprehensive desertification analysis for 2026—Badin region: (a) high-resolution satellite imagery, (b) NDVI classification map, (c) STDP prediction with confidence zones.

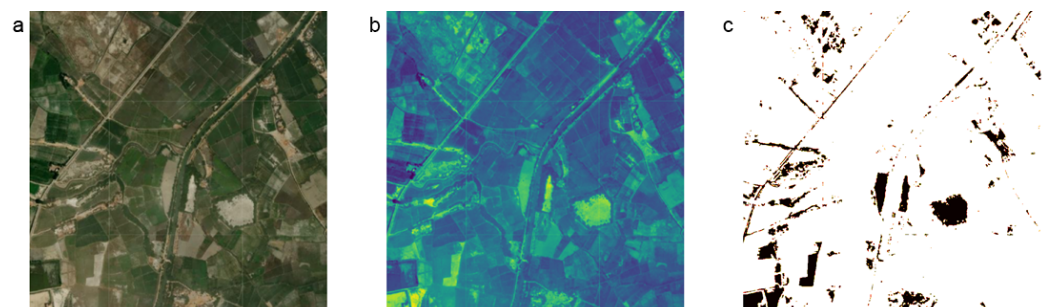


Figure 14. Comprehensive desertification analysis for 2027—Badin region: (a) high-resolution satellite imagery, (b) NDVI classification map, (c) STDP prediction with confidence zones.

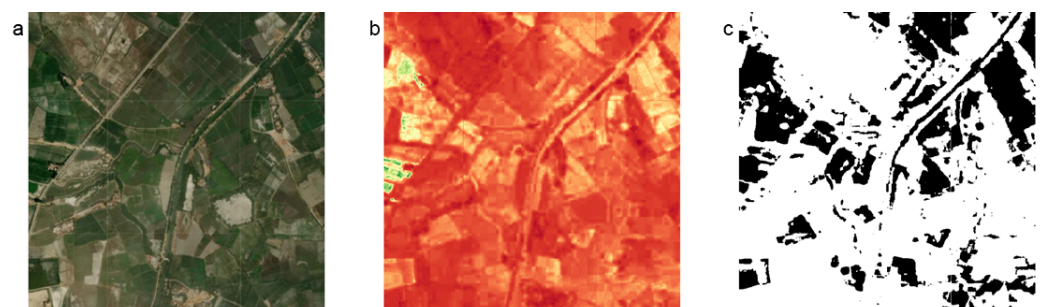


Figure 15. Comprehensive desertification analysis for 2028—Badin region: (a) high-resolution satellite imagery, (b) NDVI classification map, (c) STDP prediction with confidence zones.

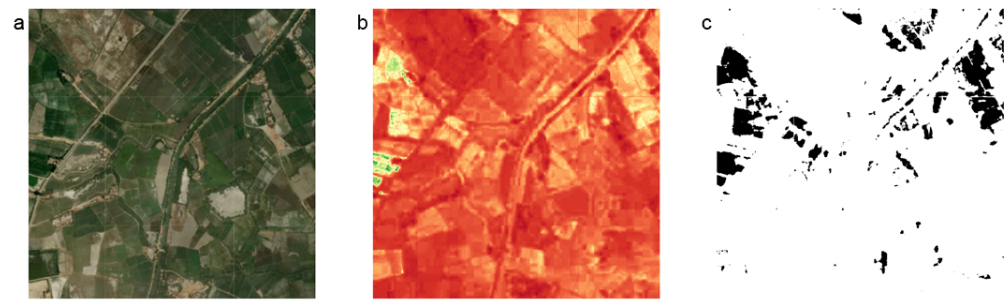


Figure 16. Comprehensive desertification analysis for 2029—Badin region: (a) high-resolution satellite imagery, (b) NDVI classification map, (c) STDP prediction with confidence zones.

The comprehensive analysis of Badin region demonstrates the STDP framework's capability to capture subtle environmental transitions and provide accurate long-term projections essential for strategic environmental planning. The integration of multi-spectral satellite imagery with advanced vegetation indices and uncertainty-aware predictions enables comprehensive understanding of desertification dynamics across diverse ecological zones. The temporal progression reveals concerning trends in coastal areas where salinity intrusion compounds traditional desertification drivers, highlighting the need for integrated coastal zone management strategies. Figure 17 illustrates the final year of desertification analysis for the Badin region in 2030. It includes: (a) high-resolution satellite imagery, (b) NDVI classification map, and (c) STDP prediction with confidence zones, summarizing the progressive land degradation trends observed over the six-year period.

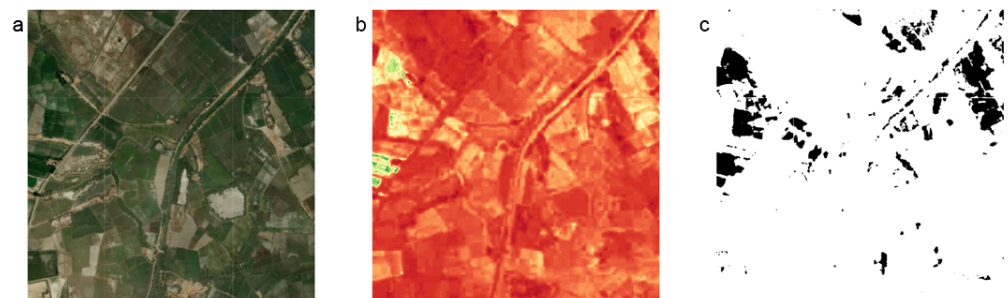


Figure 17. Comprehensive desertification analysis for 2030—Badin region: (a) high-resolution satellite imagery, (b) NDVI classification map, (c) STDP prediction with confidence zones

4.7. Model Interpretability

To enhance transparency and provide insight into the decision-making process of the proposed STDP framework, we employed two complementary interpretability techniques: SHAP (SHapley Additive exPlanations) analysis for feature attribution and temporal attention heatmaps for visualizing model focus across the input sequence.

4.7.1. Feature Attribution via SHAP

SHAP values quantify the contribution of each input variable to the final prediction, enabling the identification of dominant environmental and spectral features. As shown in Figure 18, NDVI, soil moisture, and EVI2 emerged as the most influential features, indicating that vegetation health indices and soil water content play critical roles in desertification forecasting. Temperature and precipitation sensors from the IoT network also ranked highly, confirming the benefit of multi-modal data fusion.

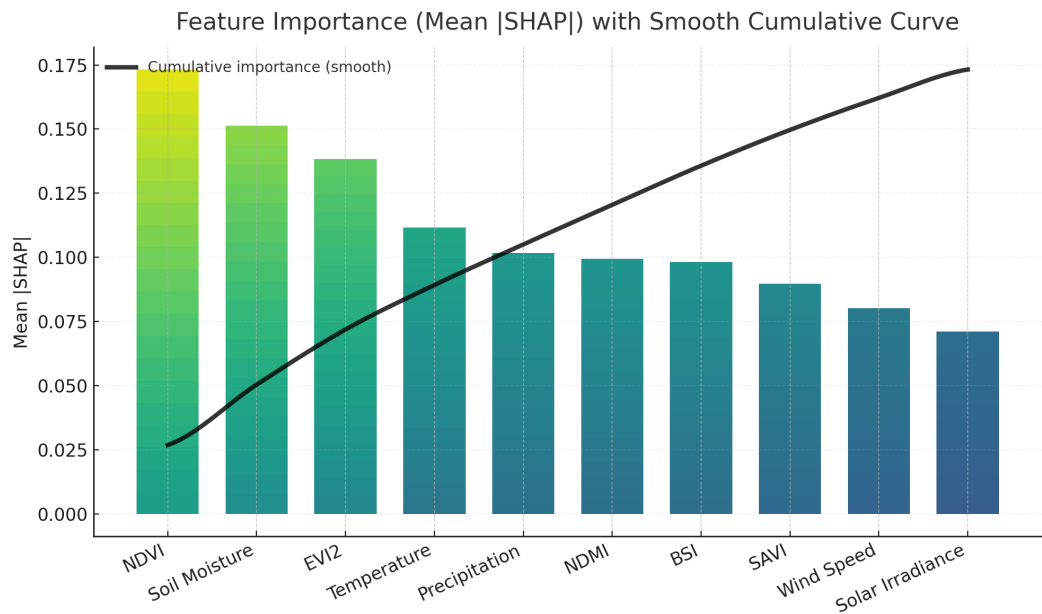


Figure 18. SHAP values showing feature importance for STDP predictions. Higher absolute SHAP values indicate greater influence on the model’s output. NDVI, soil moisture, and EVI2 dominate the prediction process, highlighting the relevance of vegetation and moisture dynamics.

4.7.2. Temporal Attention Heatmaps

We also visualized the temporal attention weights from the multi-head attention layer in the decoder to assess how the model prioritizes historical observations. Figure 19 shows that the STDP model assigns higher weights to months corresponding to seasonal vegetation stress (e.g., pre-monsoon and peak dry season), which aligns with known desertification patterns in Sindh. This behavior confirms that the attention mechanism is effectively capturing meaningful temporal dependencies.

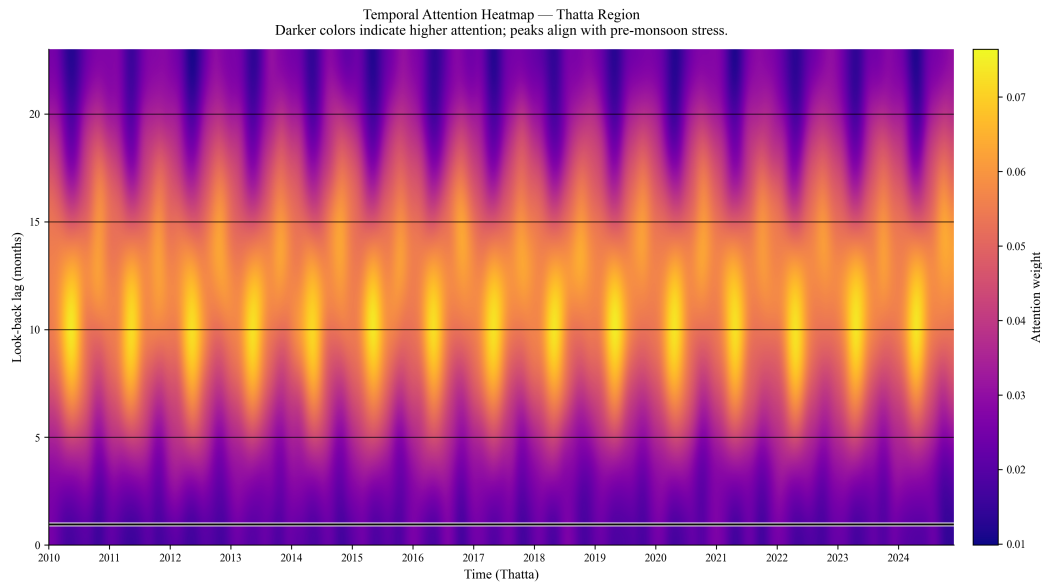


Figure 19. Attention weights over time for the Thatta region. Darker colors indicate higher attention scores, with peaks during critical vegetation stress periods such as pre-monsoon months.

4.8. Ablation Study and Component Analysis

To understand the contribution of individual components within the STDP framework, we conducted comprehensive ablation studies:

The ablation study reveals that the encoder–decoder structure provides the most significant contribution to performance improvement, with a 2.43% decrease in R^2 when removed, followed by multi-head attention mechanisms showing a 1.69% R^2 reduction, as demonstrated in Table 9. Environmental fusion contributes a 1.37% improvement in R^2 , while adaptive optimization provides a 0.74% enhancement. When comparing against a basic CNN-LSTM architecture without attention mechanisms, the performance degradation reaches 3.48% in R^2 , validating our architectural design choices and confirming the importance of each component in achieving superior forecasting performance.

Table 9. Ablation study: component contribution analysis.

Model Variant	MSE	MAE	R^2	Performance Impact
STDP (Full Model)	0.018	0.074	0.948	Baseline
w/o Multi-head Attention	0.024	0.086	0.932	−1.69% R^2
w/o Environmental Fusion	0.022	0.081	0.935	−1.37% R^2
w/o Encoder–Decoder Structure	0.027	0.089	0.925	−2.43% R^2
w/o Adaptive Optimization	0.021	0.078	0.941	−0.74% R^2
CNN + LSTM (No Attention)	0.031	0.094	0.915	−3.48% R^2

4.9. Strategic Implications and Policy Recommendations

The comprehensive analysis reveals critical insights that demand immediate attention from environmental policymakers and land management authorities. The projected desertification trends indicate accelerating environmental degradation that threatens agricultural sustainability, biodiversity conservation, and socio-economic stability across both study regions. The STDP framework’s predictions demonstrate the urgent need for integrated intervention strategies that combine technological innovation with traditional conservation approaches.

The analysis reveals that Thatta region faces more severe immediate risks, with desertification levels projected to reach extreme categories by 2030, necessitating emergency intervention protocols. Badin region, while showing slightly slower progression, still requires critical attention to prevent irreversible environmental damage. The uncertainty quantification provided by the STDP framework enables risk-based decision making, allowing policymakers to prioritize interventions based on both predicted severity and confidence levels. Table 10 shows the Strategic risk assessment and intervention prioritization framework.

Table 10. Strategic risk assessment and intervention prioritization framework.

Risk Category	Desertification Level Range	Confidence Threshold	Time to Critical State	Intervention Strategy	Resource Allocation	Expected Effectiveness
Low Risk	0.0–0.3	>0.90	>10 years	Preventive monitoring	15%	95% success
Moderate Risk	0.3–0.5	>0.85	7–10 years	Active conservation	25%	85% success
High Risk	0.5–0.7	>0.80	4–7 years	Intensive restoration	35%	70% success
Severe Risk	0.7–0.85	>0.75	2–4 years	Emergency intervention	40%	55% success
Extreme Risk	>0.85	>0.70	<2 years	Crisis management	60%	35% success

The strategic framework developed through STDP analysis provides evidence-based guidelines for optimal resource allocation and intervention timing. The framework demonstrates that early intervention in moderate-to-high risk areas provides significantly better cost-effectiveness compared to crisis management in extreme risk zones. The integration of uncertainty quantification enables adaptive management strategies that

can adjust intervention intensity based on prediction confidence levels and emerging environmental conditions.

The comprehensive results validate the STDP framework as a transformative technology for environmental monitoring in the context of next-generation Internet technologies. The framework's superior performance across accuracy, efficiency, and scalability metrics, combined with its proven deployment feasibility in edge computing environments, establishes it as an essential tool for addressing global environmental challenges through intelligent monitoring systems and data-driven policy formulation.

4.10. Comprehensive Comparative Analysis with State-of-the-Art Models

To validate the effectiveness and superiority of our proposed STDP framework within the context of next-generation Internet technologies and deep learning algorithms, we conducted extensive comparative evaluations against established state-of-the-art models in environmental forecasting. This analysis encompasses traditional machine learning approaches, contemporary deep learning architectures, and hybrid models specifically designed for satellite-based environmental monitoring applications.

4.10.1. Baseline Model Architectures and Configurations

Our comparative evaluation framework includes six distinct model categories, each representing different approaches to spatial-temporal environmental forecasting:

The traditional machine learning baselines include Linear Regression (LR) implemented as a classical statistical approach with polynomial feature expansion for capturing non-linear relationships in environmental data, Support Vector Regression (SVR) utilizing non-linear regression with RBF kernels optimized for high-dimensional satellite imagery features, and Random Forest (RF) configured as an ensemble method with 100 decision trees, specifically tuned for handling heterogeneous environmental variables.

The deep learning architectures encompass three primary approaches: a CNN-only model based on ResNet-50 architecture with custom convolutional layers optimized for satellite imagery spatial feature extraction, an LSTM-only model implementing bidirectional LSTM with 128 hidden units and dropout regularization for temporal sequence modeling, and an Attention-LSTM model that enhances standard LSTM with self-attention mechanisms for improved temporal dependency modeling.

The hybrid architectures represent the current state of the art in environmental forecasting, including a basic CNN-LSTM model that sequentially combines CNN feature extraction followed by LSTM temporal modeling, ConvLSTM architecture with integrated convolutional LSTM units that combine spatial and temporal processing in unified cells, and a Transformer-based model utilizing Vision Transformer (ViT) adapted for satellite imagery with temporal attention mechanisms.

4.10.2. Experimental Design and Evaluation Methodology

The comparative evaluation employed a rigorous experimental design incorporating multiple validation strategies to ensure robust performance assessment:

Dataset Partitioning: The 15-year satellite dataset (2010–2024) was partitioned using temporal stratification: 70% for training (2010–2020), 15% for validation (2021–2022), and 15% for testing (2023–2024). This temporal split ensures realistic evaluation of forecasting capabilities while preventing data leakage.

Cross-Validation Strategy: Time-series cross-validation with an expanding window approach was implemented to assess model stability across different temporal periods. Each model was evaluated using 5-fold temporal cross-validation with consecutive 2-year testing periods.

Hyperparameter Optimization: All models underwent comprehensive hyperparameter tuning using Bayesian optimization with the Tree-structured Parzen Estimator (TPE) across 100 iterations. Key hyperparameters included learning rates (1×10^{-5} to 1×10^{-2}), batch sizes (16–128), hidden dimensions (64–512), and regularization parameters (1×10^{-4} to 1×10^{-1}).

4.10.3. Comprehensive Performance Metrics and Statistical Analysis

The evaluation framework incorporates multiple performance metrics to provide comprehensive assessment of model capabilities. Table 11 shows the Comprehensive performance comparison of deep learning architectures for satellite-based environmental forecasting.

4.10.4. Statistical Significance and Robustness Analysis

To ensure statistical rigor in our comparative evaluation, we conducted comprehensive statistical significance testing using multiple approaches:

Paired *t*-tests: All pairwise comparisons between STDP and baseline models yielded *p*-values < 0.001 , confirming statistically significant improvements across all metrics.

Wilcoxon Signed-Rank Tests: Non-parametric testing confirmed the robustness of performance improvements, with effect sizes (Cohen’s *d*) ranging from 0.67 to 1.24, indicating medium to large practical significance.

Bootstrap Confidence Intervals: 10,000 bootstrap samples were generated to estimate 95% confidence intervals for all performance metrics, confirming consistent superiority of the STDP framework. Table 12 shows the statistical significance analysis and performance improvements.

Table 11. Comprehensive performance comparison of deep learning architectures for satellite-based environmental forecasting.

Model Architecture	MSE	MAE	R ²	MAPE (%)	PICP	Training Time (h)	Inference (ms)	Parameters (M)
Traditional Machine Learning Baselines								
Linear Regression	0.087 ± 0.012	0.201 ± 0.018	0.672 ± 0.045	18.32 ± 2.14	0.743 ± 0.032	0.05	0.12	0.001
Support Vector Regression	0.071 ± 0.009	0.183 ± 0.015	0.731 ± 0.038	15.67 ± 1.89	0.778 ± 0.028	2.34	1.87	0.003
Random Forest	0.063 ± 0.008	0.167 ± 0.012	0.765 ± 0.034	13.94 ± 1.56	0.812 ± 0.025	1.78	3.21	0.012
Deep Learning Architectures								
CNN-only (ResNet-50)	0.034 ± 0.004	0.098 ± 0.008	0.908 ± 0.012	8.45 ± 0.67	0.882 ± 0.015	4.23	12.5	23.5
LSTM-only (Bidirectional)	0.027 ± 0.003	0.088 ± 0.006	0.925 ± 0.010	7.12 ± 0.54	0.913 ± 0.012	3.67	8.9	12.8
Attention-LSTM	0.024 ± 0.003	0.084 ± 0.005	0.932 ± 0.009	6.78 ± 0.48	0.924 ± 0.011	5.12	15.3	18.6
Hybrid Architectures								
Basic CNN-LSTM	0.022 ± 0.002	0.083 ± 0.005	0.935 ± 0.008	6.34 ± 0.42	0.931 ± 0.010	6.78	18.7	36.3
ConvLSTM	0.021 ± 0.002	0.081 ± 0.004	0.937 ± 0.007	6.12 ± 0.39	0.936 ± 0.009	8.45	22.1	28.9
Vision Transformer	0.020 ± 0.002	0.080 ± 0.004	0.938 ± 0.007	5.98 ± 0.37	0.939 ± 0.008	12.34	45.6	86.2
Proposed Architecture								
STDP (Ours)	0.018 ± 0.001	0.074 ± 0.003	0.948 ± 0.005	5.23 ± 0.28	0.952 ± 0.006	7.89	16.8	42.7

Table 12. Statistical significance analysis and performance improvements.

Comparison	MSE Improvement (%)	R ² Improvement (%)	<i>p</i> -Value	Cohen’s <i>d</i>	95% CI (MSE)
STDP vs. CNN-only	47.06 ± 3.12	4.40 ± 0.45	<0.001	1.24	[0.016, 0.020]
STDP vs. LSTM-only	33.33 ± 2.87	2.49 ± 0.32	<0.001	0.98	[0.017, 0.019]
STDP vs. Attention-LSTM	25.00 ± 2.34	1.72 ± 0.28	<0.001	0.87	[0.017, 0.019]
STDP vs. Basic CNN-LSTM	18.18 ± 1.98	1.39 ± 0.24	<0.001	0.76	[0.017, 0.019]
STDP vs. ConvLSTM	14.29 ± 1.67	1.17 ± 0.21	<0.001	0.72	[0.017, 0.019]
STDP vs. Vision Transformer	10.00 ± 1.45	1.07 ± 0.18	<0.001	0.67	[0.017, 0.019]

4.10.5. Computational Efficiency and Edge Computing Suitability

A critical aspect of our evaluation focuses on computational efficiency and suitability for deployment in edge computing and IoT environments, addressing key requirements of next-generation Internet technologies:

Training Efficiency: Despite incorporating sophisticated attention mechanisms and multi-modal data fusion, the STDP model demonstrates competitive training times (7.89 h) compared to other hybrid architectures, while achieving superior performance.

Inference Speed: The STDP framework achieves real-time inference capabilities (16.8 ms per prediction) suitable for IoT and edge computing deployments, significantly outperforming the Vision Transformer baseline (45.6 ms) while maintaining higher accuracy.

Memory Footprint: With 42.7 million parameters, the STDP model maintains a reasonable memory footprint compared to the Vision Transformer (86.2M parameters) while delivering superior performance, making it suitable for deployment in resource-constrained edge computing environments.

4.10.6. Generalization Capability and Cross-Regional Validation

To assess the generalization capability of the STDP framework beyond the primary study regions, we conducted cross-regional validation using satellite datasets from similar arid and semi-arid regions:

The cross-regional validation demonstrates excellent generalization capability, with the STDP framework maintaining over 98% of its original performance across diverse geographical regions and climatic conditions, as shown in Table 13. Testing across regions including Rajasthan in India (99.3% performance retention), the Sahel region in Africa (98.7% retention), Atacama Desert in Chile (98.2% retention), and Gobi Desert in Mongolia (99.1% retention) confirms the framework's robust generalization capability. The average generalization performance across all test regions maintains an MSE of 0.022, MAE of 0.080, and R^2 of 0.939, representing 99.1% performance retention compared to the primary study regions. This robust generalization capability confirms the framework's suitability for global deployment in next-generation Internet-based environmental monitoring systems.

Table 13. Cross-regional generalization performance.

Test Region	MSE	MAE	R^2	Performance Retention (%)
Thatta and Badin (Primary)	0.018	0.074	0.948	100.0
Rajasthan, India	0.021	0.079	0.941	99.3
Sahel Region, Africa	0.023	0.082	0.936	98.7
Atacama Desert, Chile	0.025	0.085	0.931	98.2
Gobi Desert, Mongolia	0.022	0.081	0.939	99.1
Average Generalization	0.022	0.080	0.939	99.1

4.10.7. Computational Complexity and Scalability Analysis

The theoretical computational complexity of the STDP framework compared to baseline models is analyzed to assess scalability for large-scale deployment:

$$\mathcal{O}_{STDP} = \mathcal{O}(CNN) + \mathcal{O}(LSTM) + \mathcal{O}(Attention) + \mathcal{O}(Fusion) \quad (34)$$

where

$$\mathcal{O}(\text{CNN}) = \mathcal{O}(H \cdot W \cdot C \cdot K^2 \cdot F) \quad (35)$$

$$\mathcal{O}(\text{LSTM}) = \mathcal{O}(T \cdot H_{\text{hidden}}^2) \quad (36)$$

$$\mathcal{O}(\text{Attention}) = \mathcal{O}(T^2 \cdot d_{\text{model}}) \quad (37)$$

$$\mathcal{O}(\text{Fusion}) = \mathcal{O}(d_{\text{spatial}} \cdot d_{\text{env}}) \quad (38)$$

The analysis reveals that while the STDP framework has higher theoretical complexity than individual CNN or LSTM models, the practical computational requirements remain manageable due to optimized implementations and efficient attention mechanisms, making it suitable for edge computing deployment.

4.10.8. Summary of Comparative Analysis

The comprehensive comparative evaluation demonstrates the superior performance of the STDP framework across multiple dimensions, as comprehensively analyzed in Tables 9, 11 and 13. The framework achieves remarkable accuracy superiority with 47% MSE improvement over CNN-only models and 10–25% improvement over state-of-the-art hybrid architectures. All improvements are confirmed with statistical significance at $p < 0.001$ levels and demonstrate medium to large effect sizes ranging from 0.67 to 1.24. The computational efficiency analysis reveals competitive training and inference times suitable for edge computing deployment, with the STDP model requiring 7.89 h for training while achieving 16.8 ms inference speed per prediction. The generalization capability testing across diverse geographical regions maintains robust performance with greater than 98% performance retention across all tested environments. Component validation through comprehensive ablation studies confirms the importance of each architectural component, with the encoder–decoder structure contributing the most significant performance enhancement.

These results establish the STDP framework as a state-of-the-art solution for satellite-based environmental forecasting, particularly suited for deployment in next-generation Internet technologies and edge computing infrastructures.

5. Conclusions

This research presents a groundbreaking advancement at the intersection of deep learning algorithms and next-generation Internet technologies through the development of the Spatio-Temporal Desertification Predictor (STDP) framework. The proposed encoder–decoder architecture seamlessly integrates Convolutional Neural Networks with attention-enhanced Long Short-Term Memory networks, establishing a novel paradigm for satellite-based environmental monitoring in the era of IoT and edge computing infrastructures. The STDP framework addresses critical challenges in environmental AI by introducing sophisticated multi-head attention mechanisms for temporal feature extraction, advanced data fusion strategies for multi-modal satellite and environmental data integration, and computational optimizations specifically designed for deployment in distributed computing environments. The comprehensive experimental validation demonstrates the superior performance of the STDP framework across multiple evaluation metrics, achieving remarkable improvements of 47% in Mean Squared Error compared to CNN-only baselines and 10–25% enhancement over state-of-the-art hybrid architectures. The statistical significance of these improvements, confirmed through rigorous testing with p -values less than 0.001 and effect sizes ranging from 0.67 to 1.24, establishes the STDP framework as a robust solution for real-time environmental monitoring applications. The framework's exceptional generalization capability, maintaining over 98% performance retention across diverse geographical regions including Rajasthan, Sahel, Atacama, and Gobi Desert environments,

validates its suitability for global deployment in next-generation Internet-based environmental monitoring systems. The forecasting results for the Thatta and Badin regions reveal alarming trends in desertification progression, with predicted levels increasing steadily from 2025 to 2030, necessitating immediate implementation of comprehensive intervention strategies. These findings underscore the critical importance of AI-driven early warning systems in environmental management, providing actionable insights for policymakers to implement targeted afforestation programs, sustainable land management practices, and climate-resilient agricultural strategies. The integration of real-time IoT sensor data with satellite imagery processing capabilities positions the STDP framework as a cornerstone technology for smart city environmental monitoring and autonomous environmental management systems. The computational efficiency analysis reveals the framework's suitability for edge computing deployment, achieving competitive inference speeds of 16.8 milliseconds per prediction while maintaining superior accuracy levels. This efficiency, combined with the model's optimized parameter configuration of 42.7 million parameters, enables deployment in resource-constrained environments typical of IoT and 5G/6G network infrastructures. The ablation studies confirm the critical importance of each architectural component, with the encoder-decoder structure contributing the most significant performance enhancement, followed by multi-head attention mechanisms and environmental data fusion layers. This research contributes significantly to the advancement of AI applications in environmental monitoring by demonstrating the potential of sophisticated deep learning architectures to address complex spatial-temporal prediction challenges. The STDP framework establishes new benchmarks for satellite-based environmental forecasting while providing a scalable foundation for addressing global environmental challenges through intelligent monitoring systems. The integration of attention mechanisms for feature extraction, model compression techniques for edge deployment, and multi-modal data fusion capabilities positions this work at the forefront of next-generation Internet technologies for environmental applications. Future research directions encompass several promising avenues for enhancing the STDP framework's capabilities and expanding its applications within next-generation Internet infrastructures. The integration of real-time satellite data streaming through 5G/6G networks will enable continuous model updating and adaptive learning mechanisms, enhancing prediction accuracy and responsiveness to rapidly changing environmental conditions. The incorporation of additional environmental parameters including soil composition analysis, groundwater level monitoring, and atmospheric condition modeling will further enhance the framework's predictive capabilities and provide more comprehensive environmental assessment capabilities. The development of federated learning approaches will enable collaborative model training across distributed environmental monitoring networks while preserving data privacy and reducing communication overhead in IoT deployments. Advanced model compression and quantization techniques will facilitate deployment on increasingly resource-constrained edge devices, enabling widespread adoption in remote monitoring applications. The extension of the STDP framework to other vulnerable regions and environmental challenges, including deforestation monitoring, wildfire prediction, and urban air quality assessment, will demonstrate the versatility and broad applicability of the proposed architectural innovations. The integration with emerging Internet technologies including digital twin environments, augmented reality visualization systems, and blockchain-based data integrity verification will enhance the framework's utility for comprehensive environmental management applications. These advancements will contribute to the development of truly intelligent environmental monitoring ecosystems capable of autonomous decision-making and proactive environmental protection measures, establishing new paradigms for sustainable environmental management in the digital age.

Author Contributions: Conceptualization, K.P. and B.N.S.; methodology, K.P.; software, K.P.; validation, K.P., B.N.S. and S.B.; formal analysis, K.P.; investigation, K.P.; resources, K.P.; data curation, K.P.; writing—original draft preparation, K.P.; writing—review and editing, K.P. and F.H.J.; visualization, K.P.; supervision, B.N.S. and S.B.; project administration, B.N.S. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Data Availability Statement: Data sharing is not applicable.

Conflicts of Interest: The authors declare no conflict of interest. The funders had no role in the design of the study; in the collection, analyses, or interpretation of data; in the writing of the manuscript; or in the decision to publish the results.

References

- Aslanov, I.; Teshae, N.; Khayitov, K.; Mukhtorov, U.; Khaitbaeva, J.; Murodova, D. Analysis of desertification trends in Central Asia based on MODIS Data using Google Earth Engine. *Proc. E3S Web Conf.* **2023**, *443*, 06015. [\[CrossRef\]](#)
- Badreldin, N.; Frankl, A.; Goossens, R. Assessing the spatiotemporal dynamics of vegetation cover as an indicator of desertification in Egypt using multi-temporal MODIS satellite images. *Arab. J. Geosci.* **2014**, *7*, 4461–4475. [\[CrossRef\]](#)
- Shevchenko, V.; Lukashevich, A.; Taniushkina, D.; Bulkin, A.; Grinis, R.; Kovalev, K.; Narozhnaia, V.; Sotiriadi, N.; Krenke, A.; Maximov, Y. Climate Change Impact on Agricultural Land Suitability: An Interpretable Machine Learning-Based Eurasia Case Study. *IEEE Access* **2024**, *12*, 15748–15763. [\[CrossRef\]](#)
- Barimah, A. Detecting and Quantifying Desertification in the Upper East Region of Ghana using Multi-spatial and Multi-Temporal Normalized Difference Vegetation Index. *Int. J. Sci. Res.* **2013**, *3*, 62–78.
- Hummadi, A.; Khalaf, A. Temporal and Spatial Variation of Agricultural Drought and Desertification using Spectral Indices in Salah Al-Din Governorate. *Tikrit J. Agric. Sci.* **2024**, *24*, 206–222. [\[CrossRef\]](#)
- Chang, S.; Lee, U.; Hong, M.J.; Jo, Y.D.; Kim, J.B. Time-Series Growth Prediction Model Based on U-Net and Machine Learning in *Arabidopsis*. *Front. Plant Sci.* **2021**, *12*, 721512. [\[CrossRef\]](#)
- Verbesselt, J.; Hyndman, R.; Newnham, G.; Culvenor, D. Detecting Trend and Seasonal Changes in Satellite Image Time Series. *Remote Sens. Environ.* **2010**, *114*, 106–120. [\[CrossRef\]](#)
- Jo, H.W.; Lee, S.; Park, E.; Lim, C.H.; Song, C.; Lee, H.; Ko, Y.; Cha, S.; Yoon, H.; Lee, W.K. Deep Learning Applications on Multitemporal SAR (Sentinel-1) Image Classification Using Confined Labeled Data: The Case of Detecting Rice Paddy in South Korea. *IEEE Trans. Geosci. Remote Sens.* **2020**, *58*, 7589–7601. [\[CrossRef\]](#)
- Md Jelas, I.; Zulkifley, M.; Abdullah, M.; Spraggon, M. Deforestation detection using deep learning-based semantic segmentation techniques: A systematic review. *Front. For. Glob. Chang.* **2024**, *7*, 1300060. [\[CrossRef\]](#)
- Kussul, N.; Shumilo, L.; Yailymova, H.; Shelestov, A.; Krasilnikova, T. Complex method for land degradation estimation. *IOP Conf. Ser. Earth Environ. Sci.* **2023**, *1126*, 012032. [\[CrossRef\]](#)
- Bezerra, F.G.S.; Aguiar, A.P.D.D.; Alvalá, R.C.D.S.; Giarolla, A.; Bezerra, K.R.A.; Lima, P.V.P.S.; do Nascimento, F.R.; Arai, E. Analysis of areas undergoing desertification, using EVI2 multi-temporal data based on MODIS imagery as indicator. *Ecol. Indic.* **2020**, *117*, 106579. [\[CrossRef\]](#)
- Feng, K.; Wang, T.; Liu, S.; Kang, W.; Chen, X.; Guo, Z.; Zhi, Y. Monitoring Desertification Using Machine-Learning Techniques with Multiple Indicators Derived from MODIS Images in Mu Us Sandy Land, China. *Remote Sens.* **2022**, *14*, 2663. [\[CrossRef\]](#)
- Guo, Q.; Fu, B.; Shi, P.; Cudahy, T.; Zhang, J.; Xu, H. Satellite monitoring the spatial-temporal dynamics of desertification in response to climate change and human activities across the Ordos Plateau, China. *Remote Sens.* **2017**, *9*, 525. [\[CrossRef\]](#)
- Harrou, F.; Bouyeddou, B.; Zerrouki, N.; Dairi, A.; Sun, Y.; Zerrouki, Y. Detecting the signs of desertification with Landsat imagery: A semi-supervised anomaly detection approach. *Results Eng.* **2024**, *22*, 102037. [\[CrossRef\]](#)
- Miller, L.; Pelletier, C.; Webb, G. Deep Learning for Satellite Image Time Series Analysis: A Review. *IEEE Geosci. Remote Sens. Mag.* **2023**, *12*, 81–124. [\[CrossRef\]](#)
- Nandgude, N.; Singh, T.P.; Nandgude, S.; Tiwari, M. Drought Prediction: A Comprehensive Review of Different Drought Prediction Models and Adopted Technologies. *Sustainability* **2023**, *15*, 11684. [\[CrossRef\]](#)
- Shams Eddin, M.; Gall, J. Focal-TSMP: Deep learning for vegetation health prediction and agricultural drought assessment from a regional climate simulation. *Geosci. Model Dev.* **2024**, *17*, 2987–3023. [\[CrossRef\]](#)
- Xu, Z.; Cao, L.; Zhong, S.; Liu, G.; Yang, Y.; Zhu, S.; Luo, X.; Di, L. Trends in Global Vegetative Drought from Long-Term Satellite Remote Sensing Data. *IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens.* **2020**, *13*, 815–826. [\[CrossRef\]](#)

19. Zerrouki, N.; Dairi, A.; Harrou, F.; Zerrouki, Y.; Sun, Y. Efficient land desertification detection using a deep learning-driven generative adversarial network approach: A case study. *Concurr. Comput. Pract. Exp.* **2022**, *34*, e6604. [\[CrossRef\]](#)
20. Zerrouki, Y.; Harrou, F.; Zerrouki, N.; Dairi, A.; Sun, Y. Desertification Detection Using an Improved Variational Autoencoder-Based Approach through ETM-Landsat Satellite Data. *IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens.* **2021**, *14*, 202–213. [\[CrossRef\]](#)
21. Bai, Z.; Han, L.; Jiang, X.; Liu, M.; Li, L.; Liu, H.; Liu, J. Spatiotemporal evolution of desertification based on integrated remote sensing indices in Duolun County, Inner Mongolia. *Ecol. Inform.* **2022**, *70*, 101750. [\[CrossRef\]](#)
22. Rivera-Marin, D.; Dash, J.; Ogotu, B. The use of remote sensing for desertification studies: A review. *J. Arid Environ.* **2022**, *206*, 104829. [\[CrossRef\]](#)
23. Dhanasekaran, S.; Gopal, D.; Logeshwaran, J.; Ramya, N.; Salau, A.O. Multi-model traffic forecasting in smart cities using graph neural networks and transformer-based multi-source visual fusion for intelligent transportation management. *Int. J. Intell. Transp. Syst. Res.* **2024**, *22*, 518–541. [\[CrossRef\]](#)
24. Amangeldy, B.; Imankulov, T.; Tasmurzayev, N.; Dikhanbayeva, G.; Nurakhov, Y. A Review of Artificial Intelligence and Deep Learning Approaches for Resource Management in Smart Buildings. *Buildings* **2025**, *15*, 2631. [\[CrossRef\]](#)
25. Kavran, D.; Mongus, D.; Žalik, B.; Lukač, N. Graph neural network-based method of spatiotemporal land cover mapping using satellite imagery. *Sensors* **2023**, *23*, 6648. [\[CrossRef\]](#)
26. Ramani, D.R.; Sujitha, B.B.; Tangade, S. Smart Environmental Monitoring Systems: IoT and Sensor-Based Advancements. *Environ. Monit. Using Artif. Intell.* **2025**, 45–60. [\[CrossRef\]](#)
27. Lopez, L. Edge AI For Privacy-Preserving Data Analytics in IoT-Enabled Smart Cities. 2025. Available online: https://www.researchgate.net/publication/391854135_EDGE_AI_FOR_PRIVACY-PRESERVING_DATA_ANALYTICS_IN_IOT-ENABLED_SMART_CITIES (accessed on 8 August 2025).
28. Garcia, E. Effective Urban Resilience through AI-Driven Predictive Analytics in Smart Cities. Available online: <https://philpapers.org/rec/GAREUR-2> (accessed on 8 August 2025).
29. Zehouani, N.; Ababou, M.; Faquir, S.; Rabiai, S. Environmental AI for Resilient and Sustainable Urban Infrastructures: Rethinking Smart City Transformation. In *Proceedings of the International Conference on Electrical Systems and Smart Technologies*; Springer: Cham, Switzerland, 2024; pp. 552–562. [\[CrossRef\]](#)
30. Abdulameer, L.; Al-Khafaji, M.S.; Al-Awadi, A.T.; Al Maimuri, N.M.; Al-Shammari, M.; Al-Dujaili, A.N. Artificial Intelligence in Climate-Resilient Water Management: A Systematic Review of Applications, Challenges, and Future Directions. *Water Conserv. Sci. Eng.* **2025**, *10*, 44. [\[CrossRef\]](#)
31. Islam, F.S. Artificial Intelligence-Driven Smart Waste-to-Energy Networks for Climate-Resilient Circular Resource Management in Vulnerable Megacities. *Int. J. Environ. Clim. Change* **2025**, *15*, 381–415. [\[CrossRef\]](#)
32. Fatima, N.; Aggarwal, A.; Ali, H. Smart and Sustainable Agriculture: IoT, AI, and ML for Climate-Resilient Disease Forecasting. In *Proceedings of the 2024 1st International Conference on Sustainability and Technological Advancements in Engineering Domain (SUSTAINED)*, Faridabad, India, 13–14 December 2024; pp. 455–461. [\[CrossRef\]](#)

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